

Online Principal Components Analysis

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Abstract

We consider the *online* version of the well known Principal Component Analysis (PCA) problem. In standard PCA, the input to the problem is a set of d -dimensional vectors $\mathbf{X} = [\mathbf{x}_1, \dots, \mathbf{x}_n]$ and a target dimension $k < d$; the output is a set of k -dimensional vectors $\mathbf{Y} = [\mathbf{y}_1, \dots, \mathbf{y}_n]$ that minimize the reconstruction error: $\min_{\Phi} \sum_i \|\mathbf{x}_i - \Phi \mathbf{y}_i\|_2^2$. Here, $\Phi \in \mathbb{R}^{d \times k}$ is restricted to being isometric. The global minimum of this quantity, OPT_k , is obtainable by offline PCA.

In online PCA (OPCA) the setting is identical except for two differences: *i*) the vectors $\mathbf{x}_t$ are presented to the algorithm one by one and for every presented $\mathbf{x}_t$ the algorithm *must* output a vector $\mathbf{y}_t$ before receiving $\mathbf{x}_{t+1}$; *ii*) the output vectors $\mathbf{y}_t$ are ℓ dimensional with $\ell \geq k$ to compensate for the handicap of operating online. To the best of our knowledge, this paper is the first to consider this setting of OPCA. Our algorithm produces $\mathbf{y}_t \in \mathbb{R}^\ell$ with $\ell = O(k \cdot \text{poly}(1/\varepsilon))$ such that $\text{ALG} \leq \text{OPT}_k + \varepsilon \|\mathbf{X}\|_F^2$.

1 Introduction

Principal Component Analysis (PCA) is one of the most well known and widely used methods in scientific computing. It is used for dimension reduction, signal denoising, regression, correlation analysis, visualization etc. [8]. It can be described in many ways, but one that is particularly appealing in the context of online algorithms is the following. Given n high-dimensional vectors $\mathbf{x}_1, \dots, \mathbf{x}_n \in \mathbb{R}^d$ and a target dimension $k < d$, produce n low-dimensional vectors $\mathbf{y}_1, \dots, \mathbf{y}_n \in \mathbb{R}^k$ such that the reconstruction error is minimized. To define the reconstruction error, let $\mathcal{O}_{d,k}$ denote the set of $d \times k$ isometric embedding matrices:

$$\mathcal{O}_{d,k} = \{\Phi \in \mathbb{R}^{d \times k} \mid \forall \mathbf{y} \in \mathbb{R}^k \quad \|\Phi \mathbf{y}\|_2 = \|\mathbf{y}\|_2\}.$$

Then, the PCA reconstruction error is:

$$(1.1) \quad \min_{\Phi \in \mathcal{O}_{d,k}} \sum_{t=1}^n \|\mathbf{x}_t - \Phi \mathbf{y}_t\|_2^2.$$

PCA can be cast as an optimization problem. Given $\mathbf{x}_1, \dots, \mathbf{x}_n \in \mathbb{R}^d$ and $k < d$, the PCA problem is: find $\mathbf{y}_1, \dots, \mathbf{y}_n \in \mathbb{R}^k$ that are the optimal solution to the problem:

$$(1.2) \quad \min_{\mathbf{y}_t \in \mathbb{R}^k} \left(\min_{\Phi \in \mathcal{O}_{d,k}} \sum_{t=1}^n \|\mathbf{x}_t - \Phi \mathbf{y}_t\|_2^2 \right).$$

The solution to this problem goes through computing the Singular Value Decomposition (SVD) of the matrix $\mathbf{X} = [\mathbf{x}_1, \dots, \mathbf{x}_n] \in \mathbb{R}^{d \times n}$. Throughout the paper we assume that $d < n$ and $\text{rank}(\mathbf{X}) = d$. Let $\mathbf{U}_k \in \mathbb{R}^{d \times k}$ contain only the $k < d$ left singular vectors corresponding to the top k singular values of $\mathbf{X}$. Then, the optimal PCA vectors $\mathbf{y}_t$ are $\mathbf{y}_t = \mathbf{U}_k^\top \mathbf{x}_t$. Equivalently, if $\mathbf{Y} = [\mathbf{y}_1, \dots, \mathbf{y}_n] \in \mathbb{R}^{k \times n}$, then, $\mathbf{Y} = \mathbf{U}_k^\top \mathbf{X}$. Also, the best isometry matrix is $\Phi = \mathbf{U}_k$. We denote this optimal solution by OPT_k :

$$(1.3) \quad \text{OPT}_k := \min_{\mathbf{y}_t \in \mathbb{R}^k} \left(\min_{\Phi \in \mathcal{O}_{d,k}} \sum_{t=1}^n \|\mathbf{x}_t - \Phi \mathbf{y}_t\|_2^2 \right).$$

Computing the optimal vectors $\mathbf{y}_t = \mathbf{U}_k^\top \mathbf{x}_t$ requires several passes over the matrix $\mathbf{X}$. Power iteration based methods, for example, for computing $\mathbf{U}_k$ are memory and CPU efficient but require $\omega(1)$ passes over $\mathbf{X}$. Two passes also naively suffice; one to compute $\mathbf{X}\mathbf{X}^\top$ from which $\mathbf{U}_k$ is computed and one to generate the mapping $\mathbf{y}_t = \mathbf{U}_k^\top \mathbf{x}_t$. The bottleneck is in computing $\mathbf{X}\mathbf{X}^\top$ which demands $\Omega(d^2)$ auxiliary space (in memory) and $\Omega(d^2)$ operations per vector $\mathbf{x}_t$ (assuming they are dense). This is prohibitive even for moderate values of d . A significant amount of research went into reducing the computational overhead of obtaining a good approximation for $\mathbf{U}_k$ in one pass [9, 7, 6, 18, 17, 4, 14, 11, 13, 10]. Still, for all those algorithms, a second pass is needed to produce the reduced dimension vectors $\mathbf{y}_t$, with the exception of [4, 11, 18], which we discuss shortly.

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1.1 Online PCA In our setting of online PCA, the input, output, and cost function are identical to the offline case. The only difference is that the vectors $\mathbf{x}_t$ are presented to the algorithm one by one, and for every presented $\mathbf{x}_t$ the algorithm *must* output a vector $\mathbf{y}_t$ before receiving $\mathbf{x}_{t+1}$. Once some vector $\mathbf{y}_t$ is returned by the algorithm, it can not be changed at any later time. To compensate for the handicap of operating online, the target dimension of the algorithm, ℓ , is potentially larger than k .

This is a natural model for PCA when a downstream online algorithm is applied to $\mathbf{y}_t$. Examples include online algorithms for clustering (k -means, k -median), regression, classification (SVM, logistic regression), facility location, and k -server.

Our model departs from earlier definitions of online PCA. We shortly review three other definitions and point out the differences as well as highlight their limitations in this context.

1.1.1 Random projections Most similar to our work is the result of Sarlos [18], which uses the *random projection method* to perform online PCA. Using random projections, $\mathbf{y}_t = \mathbf{S}^\top \mathbf{x}_t$ where $\mathbf{S} \in \mathbb{R}^{d \times \ell}$ is generated randomly and independently from the data. For example, each element of $\mathbf{S}$ can be ± 1 with equal probability (Theorem 4.4 in [4]) or drawn from a normal Gaussian distribution (Theorem 10.5 in [11]). Then, with constant probability and for $\ell = \Theta(k/\varepsilon)$ $\min_{\Psi \in \mathbb{R}^{d \times \ell}} \sum_{t=1}^n \|\mathbf{x}_t - \Psi \mathbf{y}_t\|_2^2 \leq (1 + \varepsilon) \text{OPT}_k$. Here, the best reconstruction matrix is $\Psi = \mathbf{X} \mathbf{Y}^\dagger$ which is *not* an isometry in general.¹ We claim that this seemingly minute departure from our model is actually very significant.

To see this, let $\Phi \in \mathcal{O}_{d,k}$ be the optimal PCA projection for $\mathbf{X}$. Consider $\mathbf{y}_t \in \mathbb{R}^\ell$ whose first k coordinates contain $\Phi^\top \mathbf{x}_t$ and the remaining $\ell - k$ coordinates contain an arbitrary vector $\mathbf{z}_t \in \mathbb{R}^{\ell-k}$. In the case where $\|\mathbf{z}_t\|_2 \gg \|\Phi^\top \mathbf{x}_t\|_2$ the geometric arrangement of $\mathbf{y}_t$ potentially shares very little with that of the signal in $\mathbf{x}_t$. Yet,

$$\min_{\Psi \in \mathbb{R}^{d \times \ell}} \sum_{t=1}^n \|\mathbf{x}_t - \Psi \mathbf{y}_t\|_2^2 = \text{OPT}_k,$$

by setting $\Psi = (\Phi | \mathbf{0}_{d \times (\ell-k)})$. This would have been impossible had Ψ been restricted to being an isometry.

1.1.2 Regret minimization A regret minimization approach to online PCA was investigated in [19, 16]. In their setting of online PCA, at time t , *before* receiving

¹The notation $\mathbf{Y}^\dagger$ stands for the Moore-Penrose inverse or pseudo inverse of $\mathbf{Y}$.

the vector $\mathbf{x}_t$, the algorithm produces a rank k projection matrix $\mathbf{P}_t \in \mathbb{R}^{d \times d}$.² The authors present two methods for computing projections $\mathbf{P}_t$ such that the quantity $\sum_t \|\mathbf{x}_t - \mathbf{P}_t^\top \mathbf{x}_t\|_2^2$ converges to OPT_k in the usual no-regret sense. Since each $\mathbf{P}_t$ can be written as $\mathbf{P}_t = \mathbf{U}_t \mathbf{U}_t^\top$, for $\mathbf{U}_t \in \mathcal{O}_{d,k}$, it would seem that setting $\mathbf{y}_t = \mathbf{U}_t^\top \mathbf{x}_t$ should solve our problem. Alas, the decomposition $\mathbf{P}_t = \mathbf{U}_t \mathbf{U}_t^\top$ (and therefore $\mathbf{y}_t$) is underdetermined. Even if we ignore this issue, this objective is problematic for the sake of dimension reduction. Consider our setting where we can observe $\mathbf{x}_t$ before outputting $\mathbf{y}_t$. One can simply choose the rank 1 projection $\mathbf{P}_t = \mathbf{x}_t \mathbf{x}_t^\top / \|\mathbf{x}_t\|_2^2$. On the one hand this gives $\sum_t \|\mathbf{x}_t - \mathbf{P}_t \mathbf{x}_t\|_2^2 = 0$. On the other, it clearly does not provide meaningful dimension reduction.

1.1.3 Stochastic Setting There are three recent results [2, 15, 3] that efficiently approximate the PCA objective in Equation (1.1). They assume the input vectors $\mathbf{x}_t$ are drawn i.i.d. from a fixed (and unknown) distribution. In this setting, after observing n_0 columns $\mathbf{x}_t$ one can efficiently compute $\mathbf{U}_{n_0} \in \mathcal{O}_{d,k}$ such that it approximately spans the top k singular vectors of $\mathbf{X}$. Returning $\mathbf{y}_t = \mathbf{0}_{k \times 1}$ for $t < n_0$ and $\mathbf{y}_t = \mathbf{U}_{n_0}^\top \mathbf{x}_t$ for $t \geq n_0$ completes the algorithm. This algorithm is provably correct for some n_0 which is independent of n . This is intuitively correct but non-trivial to show, see [2, 15, 3] for more details. While the stochastic setting is very common in machine learning (e.g. the PAC model) in online systems the data distribution is *expected* to change or at least drift over time. In systems that deal with abuse detection or prevention, one can expect an almost adversarial input.

1.2 Summary of contributions Our first contribution is a deterministic online algorithm (see Algorithm 1 in Section 2) for the standard PCA objective in Eqn (1.2). Our main result (see Theorem 2.1) shows that, for any $\mathbf{X} = [\mathbf{x}_1, \dots, \mathbf{x}_n]$ in $\mathbb{R}^{d \times n}$, under some assumptions discussed below, $k < d$, and $\varepsilon > 0$, the proposed algorithm produces a set of vectors $\mathbf{y}_1, \dots, \mathbf{y}_n$ in $\mathbb{R}^\ell$ such that

$$\min_{\Phi \in \mathcal{O}_{d,\ell}} \sum_{t=1}^n \|\mathbf{x}_t - \Phi \mathbf{y}_t\|_2^2 \leq \text{OPT}_k + \varepsilon \|\mathbf{X}\|_F^2,$$

where $\ell = \lceil 8k/\varepsilon^2 \rceil$. The description of the algorithm and the proof of its correctness are given in Section 2.

While Algorithm 1 solves the main conceptual difficulty in online PCA, it has certain drawbacks:

1. It must assume that $\|\mathbf{x}_t\|_2^2 \leq \|\mathbf{X}\|_F^2 / \ell$.

²Here, $\mathbf{P}_t$ is a square projection matrix with $\mathbf{P}_t^2 = \mathbf{P}_t$

2. It requires $\|\mathbf{X}\|_F^2$ as input.
3. It spends $\Omega(d^2)$ floating point operations per $\mathbf{x}_t$ and requires auxiliary $\Theta(d^2)$ space in memory.

In section 4, we present Algorithm 2 that addresses all the issues above. That, in the cost of slightly increasing the target dimension and the additive error (see Theorem 4.1 in Section 4). We briefly explain here how we deal with the above issues:

1. We deal with arbitrary input vectors by special handling of large norm input vectors. This is a simple amendment to the algorithm which only doubles the required target dimension.
2. The new algorithm avoids requiring $\|\mathbf{X}\|_F$ as input by estimating it on the fly. A “doubling argument” analysis shows that the target dimension grows only to $\ell = O(k \log(n)/\varepsilon^2)$.³ Bounding the target dimension by $\ell = O(k/\varepsilon^3)$ requires a significant conceptual change to the algorithm.
3. The new algorithm spends only $O(d\ell)$ floating point operations per input vector and uses only $O(d\ell)$ space. It avoids storing and manipulating the covariance matrix $\mathbf{C}$ by utilizing a streaming matrix approximation technique [13].

2 A simple and inefficient OPCA algorithm

Let $\mathbf{X} = [\mathbf{x}_1, \dots, \mathbf{x}_n] \in \mathbb{R}^{d \times n}$ be the input high-dimensional vectors and $\ell < d$ be the target dimension. Algorithm 1 returns $\mathbf{Y} = [\mathbf{y}_1, \dots, \mathbf{y}_n] \in \mathbb{R}^{\ell \times n}$ that is “close” to $\mathbf{X}$ in the PCA sense (see Theorem 2.1 below for a precise statement of our quality-of-approximation result regarding Algorithm 1). Besides $\mathbf{X}$ and ℓ , the algorithm also requires $\|\mathbf{X}\|_F^2$ in its input. Moreover, we assume $\|\mathbf{x}_t\|_2^2 \leq \|\mathbf{X}\|_F^2/\ell$.

Algorithm 1 initializes $\mathbf{U}$ and $\mathbf{C}$ to $d \times \ell$ and $d \times d$ all-zeros matrices, respectively. It subsequently updates those matrices appropriately. The matrix $\mathbf{U}$ is the so-called “projection matrix”, i.e., $\mathbf{y}_t = \mathbf{U}^\top \mathbf{x}_t$. The matrix $\mathbf{C}$ is an auxiliary matrix to accumulate all the residual errors. The residual for a vector $\mathbf{x}_t$ is: $\mathbf{r}_t = \mathbf{x}_t - \mathbf{U}\mathbf{U}^\top \mathbf{x}_t$. The algorithm starts with a rank one update of $\mathbf{C}$ as $\mathbf{C} = \mathbf{C} + \mathbf{r}_1 \mathbf{r}_1^\top$. Notice that by the assumption for $\mathbf{x}_t$, we have that $\|\mathbf{r}_1 \mathbf{r}_1^\top\|_2^2 \leq \|\mathbf{X}\|_F^2/\ell$, and hence for $t = 1$ the algorithm will not enter the while-loop. Then, for the second input vector $\mathbf{x}_2$, the algorithm proceeds by checking the spectral norm of $\mathbf{C} + \mathbf{r}_2 \mathbf{r}_2^\top = \mathbf{r}_1 \mathbf{r}_1^\top + \mathbf{r}_2 \mathbf{r}_2^\top$. If this does not exceed the threshold θ , which is always fixed to $\theta = 2 \cdot \|\mathbf{X}\|_F^2/\ell$, the algorithm keeps $\mathbf{U}$ unchanged, and it can go all the way

Algorithm 1 An online algorithm for Principal Component Analysis

Input: $\mathbf{X} = [\mathbf{x}_1, \dots, \mathbf{x}_n] \in \mathbb{R}^{d \times n}$ with $\|\mathbf{x}_t\|_2^2 \leq \|\mathbf{X}\|_F^2/\ell$, $\ell < d$, $\|\mathbf{X}\|_F^2$
 $\mathbf{U} = \mathbf{0}_{d \times \ell}$;
 $\mathbf{C} = \mathbf{0}_{d \times d}$;
 $\theta = 2\|\mathbf{X}\|_F^2/\ell$;
for $t = 1, \dots, n$ **do**
 $\mathbf{r}_t = \mathbf{x}_t - \mathbf{U}\mathbf{U}^\top \mathbf{x}_t$
 while $\|\mathbf{C} + \mathbf{r}_t \mathbf{r}_t^\top\|_2 \geq \theta$ **do**
 $[\mathbf{u}, \lambda] \leftarrow \text{TopEigenvectorAndValueOf}(\mathbf{C})$
 add $\mathbf{u}$ to the next zero column of $\mathbf{U}$
 $\mathbf{r}_t \leftarrow \mathbf{x}_t - \mathbf{U}\mathbf{U}^\top \mathbf{x}_t$
 $\mathbf{C} \leftarrow \mathbf{C} - \lambda \mathbf{u} \mathbf{u}^\top$
 end while
 $\mathbf{C} \leftarrow \mathbf{C} + \mathbf{r}_t \mathbf{r}_t^\top$
 $\mathbf{y}_t \leftarrow \mathbf{U}^\top \mathbf{x}_t$
end for

to $t = n$ if this is the case for all $t > 1$. Notice, then, that the fact that $\|\mathbf{C}\|_2 \leq \theta$, implies that the spectral norm squared of $\mathbf{R} = [\mathbf{r}_1, \dots, \mathbf{r}_n] \in \mathbb{R}^{d \times n}$ is bounded by θ , because $\mathbf{C} = \mathbf{R}\mathbf{R}^\top$, and as we will see in the proof of Theorem 2.1 this is a key technical component in proving that the algorithm returns $\mathbf{y}_t$ that are close to $\mathbf{x}_t$ in the PCA sense.

If, however, for some iterate t the spectral norm of $\mathbf{C} + \mathbf{r}_t \mathbf{r}_t^\top$ exceeds the threshold θ , then, the algorithm does a “correction” to $\mathbf{U}$, consequently to $\mathbf{r}_t$, in order to ensure that this is not the case. Specifically, it updates $\mathbf{U}$ with the principal eigenvector of $\mathbf{C}$ at that point of the algorithm execution. At the same time it downdates $\mathbf{C}$ (inside the while-loop) by removing this eigenvector. That way the algorithm ensures that at the end of each iterate t , the following relation is true: $\mathbf{C} = \sum_t \mathbf{r}_t \mathbf{r}_t^\top - \sum_j \lambda_j \mathbf{u}_j \mathbf{u}_j^\top$ with $\mathbf{C}(\sum_j \lambda_j \mathbf{u}_j \mathbf{u}_j^\top) = \mathbf{0}_{d \times d}$ (for a formal statement of the latter orthogonality condition see Lemma 3.1). Hence, when the condition in the while-loop gives $\|\mathbf{C} + \mathbf{r}_t \mathbf{r}_t^\top\|_2^2 \leq \theta$, which implies that $\|\mathbf{C}\|_2^2 \leq \theta$, it really means (by an orthogonality argument - see Lemma 3.6) that $\|\sum_t \mathbf{r}_t \mathbf{r}_t^\top\|_2^2 \leq \theta$, which is the ultimate goal of the algorithm as we argued before.

2.1 Main result The following theorem is our main quality-of-approximation result regarding Algorithm 1. We prove the theorem based on several other facts which we state and prove in the next section.

THEOREM 2.1. *Let $\mathbf{X} = [\mathbf{x}_1, \dots, \mathbf{x}_n] \in \mathbb{R}^{d \times n}$ with $\|\mathbf{x}_t\|_2^2 \leq \|\mathbf{X}\|_F^2/\ell$, $\ell < d$, and $\|\mathbf{X}\|_F^2$ be inputs to Algorithm 1. For any target dimension $k < d$, and any accuracy parameter $\varepsilon > 0$, let the target dimension of the algorithm be $\ell = \lceil 8k/\varepsilon^2 \rceil < d$. Then:*

³Here, we assume that $\|\mathbf{x}_t\|$ are polynomial in n .

1. The algorithm terminates and the while-loop runs at most ℓ times (see Lemma 3.7).
2. Upon termination of the algorithm:

$$\text{ALG}_\ell := \min_{\Phi \in \mathcal{O}_{d,\ell}} \sum_{t=1}^n \|\mathbf{x}_t - \Phi \mathbf{y}_t\|_2^2 \leq \text{OPT}_k + \varepsilon \cdot \|\mathbf{X}\|_{\text{F}}^2.$$

3. The algorithm uses amortized $O(d^3)$ arithmetic operations per vector $\mathbf{x}_t$ and $O(d^2)$ auxiliary space.

Proof. We quickly prove item 2 in the theorem by using several intermediate results all proven in Section 3. Let $\mathbf{R}$ be the $d \times n$ matrix containing $\mathbf{r}_t$ in its t -th column⁴, and let $\mathbf{Y}$ be the $\ell \times n$ matrix containing $\mathbf{y}_t$ in its t -th column. In Lemma 3.4 we prove:

$$\min_{\Phi \in \mathcal{O}_{d,\ell}} \sum_{t=1}^n \|\mathbf{x}_t - \Phi \mathbf{y}_t\|_2^2 \leq \|\mathbf{R}\|_{\text{F}}^2;$$

next, in Lemma 3.5 we prove:

$$\|\mathbf{R}\|_{\text{F}}^2 \leq \text{OPT}_k + 2 \cdot \|\mathbf{X}\|_{\text{F}} \cdot \sqrt{k} \cdot \|\mathbf{R}\|_2;$$

and in Lemma 3.6 we prove:

$$\|\mathbf{R}\|_2^2 \leq 2\|\mathbf{X}\|_{\text{F}}^2/\ell.$$

Combining these three bounds gives,

$$\min_{\Phi \in \mathcal{O}_{d,\ell}} \sum_{t=1}^n \|\mathbf{x}_t - \Phi \mathbf{y}_t\|_2^2 \leq \text{OPT}_k + 2 \cdot \|\mathbf{X}\|_{\text{F}} \cdot \sqrt{\frac{2k}{\ell}} \cdot \|\mathbf{X}\|_{\text{F}}.$$

Use $\ell = \lceil 8k/\varepsilon^2 \rceil$ in the latter equation to wrap up the proof of the second item in the theorem. ■

2.2 Finding the best isometry matrix We remark that our algorithm does *not* compute the best isometry matrix Φ , for which the bound in Theorem 2.1 holds. The algorithm indeed only returns the low-dimensional vectors $\mathbf{y}_t$. The best Φ , after $\mathbf{X}$ and $\mathbf{Y}$ are fixed, is related to the so-called Procrustes problem:

$$\argmin_{\Phi \in \mathcal{O}_{d,\ell}} \|\mathbf{X} - \Phi \mathbf{Y}\|_{\text{F}}^2.$$

Let $\mathbf{X}\mathbf{Y}^T = \mathbf{U}\Sigma\mathbf{V}^T \in \mathbb{R}^{d \times \ell}$ be the SVD of $\mathbf{X}\mathbf{Y}^T$ with $\mathbf{U} \in \mathbb{R}^{d \times \ell}$, $\Sigma \in \mathbb{R}^{\ell \times \ell}$ and $\mathbf{V} \in \mathbb{R}^{\ell \times \ell}$. Then, $\Phi = \mathbf{U}\mathbf{V}^T$. Clearly, it is not possible to construct this matrix in an online fashion. However, our algorithm does find an isometry matrix (the matrix $\mathbf{U} \in \mathbb{R}^{d \times \ell}$ upon termination of the algorithm) which satisfies the bound in Theorem 2.1 (see the proof of Lemma 3.4 to verify this claim).

⁴Here, $\mathbf{r}_t$ is the corresponding vector upon termination of the corresponding iterate of the algorithm.

3 Auxiliary Lemmas

3.1 Notation We introduce notation that, though not necessary to describe the algorithm, is useful for proving our results regarding Algorithm 1.

Let ℓ' be the number of vectors $\mathbf{u}$ inserted in $\mathbf{U} \in \mathbb{R}^{d \times \ell}$. This is exactly the number of times the while-loop is executed. We argue in Lemma 3.7 that $\ell' \leq \ell$. Notice that after the execution of the algorithm $\mathbf{U}$ may still contain some all-zero columns.

We reserve the index t to refer to the iterate of the algorithm where the algorithm receives $\mathbf{x}_t$. Also, let $\mathbf{U}_t \in \mathbb{R}^{d \times \ell}$ denote the projection matrix $\mathbf{U}$ used for the vector $\mathbf{x}_t$.

We reserve the index i to index the various instances of the matrix $\mathbf{C}$ during the execution of the algorithm. Hence, i ranges from $i = 1$ ($\mathbf{C}_1 = \mathbf{0}_{d \times d}$) to $i = z$, where z is the number of times the algorithm updates $\mathbf{C}$. Clearly, $z \geq n$, because for each t there is at least one such update (outside the while-loop). Also, $z = n + \ell' \leq n + \ell$, since the while-loop is executed at most ℓ times.

We reserve the index j to index the various vectors $\mathbf{u}$ added to the matrix $\mathbf{U}$ during the execution of the algorithm. For $j = 1, 2, \dots, \ell'$, $\mathbf{u}_j \in \mathbb{R}^d$ are the vectors $\mathbf{u}$ computed inside the while-loop in Algorithm 1. Notice that $\mathbf{u}_j$ is the eigenvector of some $\mathbf{C}_i$. Also, let λ_j be the corresponding largest eigenvalue of $\mathbf{C}_i$ ($\mathbf{C}_i$ is a symmetric matrix as we argue in Eqn. (3.4)) such that $\lambda_j = \mathbf{u}_j^T \mathbf{C}_i \mathbf{u}_j$.

Let $\mathbf{X} \in \mathbb{R}^{d \times n}$, $\mathbf{Y} \in \mathbb{R}^{\ell \times n}$, $\tilde{\mathbf{X}} \in \mathbb{R}^{d \times n}$ and $\mathbf{R} \in \mathbb{R}^{d \times n}$, denote the matrices whose t 'th column is $\mathbf{x}_t \in \mathbb{R}^d$, $\mathbf{y}_t \in \mathbb{R}^\ell$, $\tilde{\mathbf{x}}_t = \mathbf{U}_t \mathbf{U}_t^T \mathbf{x}_t \in \mathbb{R}^d$, and $\mathbf{r}_t \in \mathbb{R}^d$. The vectors $\mathbf{r}_t$'s are taken as the ones in the end of the corresponding iteration.

By construction of the algorithm the following relation is also true

$$(3.4) \quad \mathbf{C}_z = \mathbf{R}\mathbf{R}^T - \sum_{j=1}^{\ell'} \lambda_j \mathbf{u}_j \mathbf{u}_j^T.$$

3.2 Results The following lemma argues about some properties of the vectors $\mathbf{u}$ and the matrix $\mathbf{C}$. It argues that the vectors $\mathbf{u}$ inserted in $\mathbf{U}$ are orthogonal to each other. This is important in interpreting our algorithm as a “true” PCA algorithm, since those vectors play the role of the left singular vectors of $\mathbf{X}$, and as such should at the very least be orthogonal to each other. The lemma also shows that the vectors $\mathbf{u}$ inserted in $\mathbf{U}$ are perpendicular to the subspace spanned by the columns of the matrix $\mathbf{C}_z$. Recall that in Eqn. (3.4) we

showed:

$$\mathbf{R}\mathbf{R}^\top = \mathbf{C}_z + \sum_{j=1}^{\ell'} \lambda_j \mathbf{u}_j \mathbf{u}_j^\top;$$

hence the lemma argues that $\mathbf{C}_z$ and $\sum_{j=1}^{\ell'} \lambda_j \mathbf{u}_j \mathbf{u}_j^\top$ span two perpendicular subspaces which combined, span the subspace spanned by $\mathbf{R}\mathbf{R}^\top$. This will be useful in bounding the spectral norm of $\mathbf{R}\mathbf{R}^\top$ in Lemma 3.6. The idea is that for such a matrix $\mathbf{R}\mathbf{R}^\top$, its spectral norm can be bounded as the maximum spectral norm of $\mathbf{C}_z$ and $\sum_{j=1}^{\ell'} \lambda_j \mathbf{u}_j \mathbf{u}_j^\top$, which in turn are bounded by design of the algorithm.

LEMMA 3.1. *Let $\mathbf{U}_t \in \mathbb{R}^{d \times \ell}$ be the instance of $\mathbf{U}$ in Algorithm 1 when calculating the corresponding vector $\mathbf{y}_t = \mathbf{U}^\top \mathbf{x}_t$. Then, for all $t = 1, \dots, n$ and for some $j \in [1, \dots, \ell']$ (not necessarily the same j for different t):*

$$\mathbf{U}_t^\top \mathbf{U}_t = \begin{pmatrix} \mathbf{I}_j & \\ & \mathbf{0}_{(\ell-j) \times (\ell-j)} \end{pmatrix} \in \mathbb{R}^{\ell \times \ell}.$$

Also, for the final value taken by the matrix $\mathbf{C}$ it holds that

$$\mathbf{C}_z \left(\sum_{j=1}^{\ell'} \lambda_j \mathbf{u}_j \mathbf{u}_j^\top \right) = \mathbf{0}_{d \times d}.$$

Proof. See Section 5.1.

The following two lemmas prove upper and lower bounds for the values λ_j calculated in the algorithm. The lower bounds are useful in upper bounding the number of times the algorithm enters the while-loop (see Lemma 3.7); the upper bounds will be useful in providing an upper bound for the spectral norm of the matrix $\mathbf{R}$ (see Lemma 3.6), which is crucial for providing the error accuracy guarantee of the algorithm in Theorem 2.1.

LEMMA 3.2. (UPPER BOUND ON λ_j 'S) *Let λ_j , for $j = 1, \dots, \ell'$ be the eigenvalues computed in Algorithm 1 with λ_j computed before λ_{j+1} . Then, for all $j = 1, 2, \dots, \ell'$:*

$$\lambda_j \leq 2\|\mathbf{X}\|_{\mathbb{F}}^2/\ell.$$

Proof. Each λ_j corresponds to the largest eigenvalue/singular value of some $\mathbf{C}_i$.⁵ Hence, it suffices to argue that for all $i = 1, \dots, z$: $\sigma_1(\mathbf{C}_i) = \|\mathbf{C}_i\|_2 \leq 2\|\mathbf{X}\|_{\mathbb{F}}^2/\ell$. We prove the result by induction on i . For $i = 1$, it is $\mathbf{C}_1 = \mathbf{0}_{d \times d}$ and it is trivial that $\sigma_1(\mathbf{C}_1) = 0 \leq 2\|\mathbf{X}\|_{\mathbb{F}}^2/\ell$. By the induction hypothesis: for some $i > 1$ let $\|\mathbf{C}_i\|_2 \leq \|\mathbf{X}\|_{\mathbb{F}}^2/\ell$. We want to show that $\|\mathbf{C}_{i+1}\|_2 \leq$

$2\|\mathbf{X}\|_{\mathbb{F}}^2/\ell$. Notice that either $\mathbf{C}_{i+1} = \mathbf{C}_i + \mathbf{r}_t \mathbf{r}_t^\top$ (for some iterate t) or $\mathbf{C}_{i+1} = \mathbf{C}_i - \lambda_j \mathbf{u}_j \mathbf{u}_j^\top$ (for some j). If $\mathbf{C}_{i+1} = \mathbf{C}_i + \mathbf{r}_t \mathbf{r}_t^\top$, then, $\|\mathbf{C}_i + \mathbf{r}_t \mathbf{r}_t^\top\|_2 \leq 2\|\mathbf{X}\|_{\mathbb{F}}^2/\ell$ is ensured by the while-loop condition in the algorithm, since that update happens outside the while-loop. If $\mathbf{C}_{i+1} = \mathbf{C}_i - \lambda_j \mathbf{u}_j \mathbf{u}_j^\top$ (inside the while-loop), then,

$$\begin{aligned} \|\mathbf{C}_{i+1}\|_2 &= \|\mathbf{C}_i - \lambda_j \mathbf{u}_j \mathbf{u}_j^\top\|_2 = \sigma_1(\mathbf{C}_i - \lambda_j \mathbf{u}_j \mathbf{u}_j^\top) = \\ (3.5) \quad &= \sigma_2(\mathbf{C}_i) \leq \sigma_1(\mathbf{C}_i) = \|\mathbf{C}_i\|_2 \leq 2\|\mathbf{X}\|_{\mathbb{F}}^2/\ell, \end{aligned}$$

where $\|\mathbf{C}_i\|_2 \leq 2\|\mathbf{X}\|_{\mathbb{F}}^2/\ell$ uses the induction hypothesis. Also, in the third equality we use the fact that $(\lambda_j, \mathbf{u}_j)$ is the top eigen-pair of $\mathbf{C}_i$ hence “removing” it from $\mathbf{C}_i$ turns the second largest eigenvalue of $\mathbf{C}_i$ to be the largest one in $\mathbf{C}_i - \lambda_j \mathbf{u}_j \mathbf{u}_j^\top$. ■

LEMMA 3.3. (LOWER BOUND ON λ_j 'S) *Let λ_j , for $j = 1, \dots, \ell'$ be the eigenvalues computed in Algorithm 1 with λ_j computed before λ_{j+1} . Assuming that for all t , $\|\mathbf{x}_t\|_2^2 \leq \|\mathbf{X}\|_{\mathbb{F}}^2/\ell$, then, for all $j = 1, 2, \dots, \ell'$:*

$$\lambda_j \geq \|\mathbf{X}\|_{\mathbb{F}}^2/\ell.$$

Proof. Each λ_j corresponds to the largest eigenvalue of some $\mathbf{C}_i$. And by design of the algorithm, the extracting of λ_j from $\mathbf{C}_i$ is done inside the while-loop. Note that the condition in the while-loop is “ $\|\mathbf{C} + \mathbf{r}_t \mathbf{r}_t^\top\|_2 \geq 2\|\mathbf{X}\|_{\mathbb{F}}^2/\ell$ ”. This implies that for the iterate t :

$$\|\mathbf{C}_i\|_2 \geq \|\mathbf{C}_i + \mathbf{r}_t \mathbf{r}_t^\top\|_2 - \|\mathbf{r}_t \mathbf{r}_t^\top\|_2 \geq \|\mathbf{X}\|_{\mathbb{F}}^2/\ell.$$

The first inequality follows by the triangle inequality: $\|\mathbf{C}_i + \mathbf{r}_t \mathbf{r}_t^\top\|_2 \leq \|\mathbf{C}_i\|_2 + \|\mathbf{r}_t \mathbf{r}_t^\top\|_2$. The second inequality uses $\|\mathbf{C}_i + \mathbf{r}_t \mathbf{r}_t^\top\|_2 \geq 2\|\mathbf{X}\|_{\mathbb{F}}^2/\ell$ and $\|\mathbf{r}_t \mathbf{r}_t^\top\|_2 \leq \|\mathbf{X}\|_{\mathbb{F}}^2/\ell$. To verify that $\|\mathbf{r}_t \mathbf{r}_t^\top\|_2 \leq \|\mathbf{X}\|_{\mathbb{F}}^2/\ell$, recall that $\mathbf{r}_t = (\mathbf{I}_d - \mathbf{U}_t \mathbf{U}_t^\top) \mathbf{x}_t$. Then, using the fact that $(\mathbf{I}_d - \mathbf{U}_t \mathbf{U}_t^\top)$ is a projection matrix: $\|\mathbf{r}_t \mathbf{r}_t^\top\|_2 = \|\mathbf{r}_t\|_2^2 \leq \|\mathbf{x}_t\|_2^2 \leq \|\mathbf{X}\|_{\mathbb{F}}^2/\ell$, where the last inequality uses the assumption in the lemma. ■

The next lemma argues that the reconstruction error of our algorithm is bounded from $\|\mathbf{R}\|_{\mathbb{F}}^2$. This result is based on the fact that the matrix $\mathbf{U}$ contains orthonormal columns and the updates on $\mathbf{U}$ occur by adding columns one after the other, without ever deleting any of them. The first inequality in the derivation in the proof of the lemma also indicates that the bound in Theorem 2.1 not only holds for the best isometry matrix Φ but also for the matrix $\mathbf{U}_n$, i.e., the matrix $\mathbf{U}$ upon termination Algorithm 1, which is also an isometry.

LEMMA 3.4. *Let $\mathbf{X} \in \mathbb{R}^{d \times n}$, $\mathbf{Y} \in \mathbb{R}^{\ell \times n}$, and $\mathbf{R} \in \mathbb{R}^{d \times n}$, be the matrices whose t 'th column is $\mathbf{x}_t \in \mathbb{R}^d$, $\mathbf{y}_t \in \mathbb{R}^\ell$,*

⁵For the purposes of this proof, $\sigma_i(\mathbf{A})$ denotes the i th largest singular value of some symmetric matrix $\mathbf{A}$.

and $\mathbf{r}_t \in \mathbb{R}^d$ (the vectors $\mathbf{r}_t$'s are taken as the ones in the end of the corresponding iteration), respectively. Then,

$$\min_{\Phi \in \mathcal{O}_{d \times \ell}} \|\mathbf{X} - \Phi \mathbf{Y}\|_F^2 \leq \|\mathbf{R}\|_F^2.$$

Proof. We manipulate the term $\min_{\Phi \in \mathcal{O}_{d \times \ell}} \|\mathbf{X} - \Phi \mathbf{Y}\|_F^2$ as follows:

$$\begin{aligned} \min_{\Phi \in \mathcal{O}_{d \times \ell}} \|\mathbf{X} - \Phi \mathbf{Y}\|_F^2 &\leq \|\mathbf{X} - \mathbf{U}_n \mathbf{Y}\|_F^2 = \\ &= \sum_{i=1}^n \|\mathbf{x}_t - \mathbf{U}_n \mathbf{U}_t^\top \mathbf{x}_t\|_2^2 = \sum_{i=1}^n \|\mathbf{x}_t - \mathbf{U}_t \mathbf{U}_t^\top \mathbf{x}_t\|_2^2 = \|\mathbf{R}\|_F^2. \end{aligned}$$

The first inequality holds because $\mathbf{U}_n$ is an isometry. This holds because $\mathbf{U}_n$ contains a subset of ℓ' orthonormal columns, according to Lemma 3.1; the remaining columns are all-zeros. Also, the bottom $\ell - \ell'$ rows in $\mathbf{Y}$ are all-zeros. In the second equality, $\mathbf{U}_n \mathbf{U}_t^\top = \mathbf{U}_t \mathbf{U}_t^\top$, since $\mathbf{U}_t = [\mathbf{u}_1, \mathbf{u}_2, \dots, \mathbf{u}_{j(t)}, \mathbf{0}, \dots, \mathbf{0}]$ for some $j(t)$ and $\mathbf{U}_n = [\mathbf{u}_1, \mathbf{u}_2, \dots, \mathbf{u}_{j(n)}, \mathbf{0}, \dots, \mathbf{0}]$ with $j(n) \geq j(t)$ and all the vectors $\mathbf{u}_j$ are orthonormal. ■

Next, we provide an upper bound for the Frobenius norm squared of $\mathbf{R}$ with respect to the spectral norm of $\mathbf{R}$. That helps because the algorithm only provides a bound for $\|\mathbf{R}\|_2$ (see Lemma 3.6).

LEMMA 3.5. *Let $\mathbf{X}, \mathbf{R} \in \mathbb{R}^{d \times n}$, be the matrices whose t 'th column is $\mathbf{x}_t$ and $\mathbf{r}_t \in \mathbb{R}^d$ (the vectors $\mathbf{r}_t$'s are taken as the ones in the end of the corresponding iteration), respectively. Then, $\|\mathbf{R}\|_F^2 \leq \text{OPT}_k + 2 \cdot \|\mathbf{X}\|_F \cdot \sqrt{k} \cdot \|\mathbf{R}\|_2$*

Proof. For notational convenience, let

$$\Delta_1 = \left(\sum_{i=1}^k (\|\mathbf{v}_i^\top \mathbf{X}\|_2^2 - 2 \langle \mathbf{v}_i^\top \mathbf{X}, \mathbf{v}_i^\top \mathbf{R} \rangle + \|\mathbf{v}_i^\top \mathbf{R}\|_2^2) \right),$$

and

$$\Delta_2 = \sqrt{\sum_{i=1}^k \|\mathbf{v}_i^\top \mathbf{X}\|_2^2} \cdot \sqrt{\sum_{i=1}^k \|\mathbf{v}_i^\top \mathbf{R}\|_2^2}.$$

We manipulate the term $\|\mathbf{R}\|_F^2$ as follows.

$$\begin{aligned} \|\mathbf{R}\|_F^2 &\stackrel{(i)}{=} \|\mathbf{X}\|_F^2 - \|\tilde{\mathbf{X}}\|_F^2 \\ &\stackrel{(ii)}{\leq} \|\mathbf{X}\|_F^2 - \sum_{i=1}^k \|\mathbf{v}_i^\top \tilde{\mathbf{X}}\|_2^2 \\ &\stackrel{(iii)}{=} \|\mathbf{X}\|_F^2 - \sum_{i=1}^k \|\mathbf{v}_i^\top \mathbf{X} - \mathbf{v}_i^\top \mathbf{R}\|_2^2 \\ &\stackrel{(iv)}{=} \|\mathbf{X}\|_F^2 - \Delta_1 \\ &\stackrel{(v)}{\leq} \|\mathbf{X}\|_F^2 \\ &\quad - \sum_{i=1}^k \|\mathbf{v}_i^\top \mathbf{X}\|_2^2 + 2 \cdot \sum_{i=1}^k \|\mathbf{v}_i^\top \mathbf{X}\|_2 \|\mathbf{v}_i^\top \mathbf{R}\|_2 \\ &\stackrel{(vi)}{\leq} \|\mathbf{X}\|_F^2 - \sum_{i=1}^k \|\mathbf{v}_i^\top \mathbf{X}\|_2^2 + 2 \cdot \Delta_2 \\ &\stackrel{(vii)}{\leq} \text{OPT}_k + 2 \cdot \|\mathbf{X}\|_F \cdot \sqrt{k} \cdot \|\mathbf{R}\|_2 \end{aligned}$$

(i) follows by Lemma 5.1 in Section 5. In (ii), for $i = 1, \dots, k$ let $\mathbf{v}_i \in \mathbb{R}^d$ denote the i th left singular vector of $\mathbf{X}$ corresponding to the i th largest singular value of $\mathbf{X}$, also let $\mathbf{V} \in \mathbb{R}^{d \times k}$ contains those vectors as columns. In this equation we used $\|\tilde{\mathbf{X}}\|_F^2 \geq \sum_{i=1}^k \|\mathbf{v}_i^\top \tilde{\mathbf{X}}\|_2^2$. This is true because :

$$\sum_{i=1}^k \|\mathbf{v}_i^\top \tilde{\mathbf{X}}\|_2^2 = \|\mathbf{V}^\top \tilde{\mathbf{X}}\|_F^2 \leq \|\mathbf{V}^\top\|_2^2 \cdot \|\tilde{\mathbf{X}}\|_F^2 = \|\tilde{\mathbf{X}}\|_F^2.$$

In the first inequality we used the fact that $\|\mathbf{X}\mathbf{Y}\|_F^2 \leq \|\mathbf{X}\|_2^2 \|\mathbf{Y}\|_F^2$, for any $\mathbf{X}$ and $\mathbf{Y}$; in the last equality we used $\|\mathbf{V}^\top\|_2^2 = 1$. In (iii) we use the relation $\mathbf{X} = \tilde{\mathbf{X}} + \mathbf{R}$. In (iv) we used the fact that for any two vectors α, β : $\|\alpha - \beta\|_2^2 = \|\alpha\|_2^2 + \|\beta\|_2^2 - 2 \langle \alpha, \beta \rangle$. We used this property k times for all $i = 1, \dots, k$, with $\alpha = \mathbf{v}_i^\top \mathbf{X}$ and $\beta = \mathbf{v}_i^\top \mathbf{R}$. In (vi) we used the Cauchy-Swartz inequality: for numbers $\gamma_i \geq 0, \delta_i \geq 0$ and for $i = 1, \dots, k$ it is: $\sum_{i=1}^k \gamma_i \delta_i \leq \sqrt{\sum_{i=1}^k \gamma_i^2} \cdot \sqrt{\sum_{i=1}^k \delta_i^2}$. We used this inequality for $\gamma_i = \|\mathbf{v}_i^\top \mathbf{X}\|_2^2$ and $\delta_i = \|\mathbf{v}_i^\top \mathbf{R}\|_2^2$. In (vii) we used that

$$\begin{aligned} \|\mathbf{X}\|_F^2 - \sum_{i=1}^k \|\mathbf{v}_i^\top \mathbf{X}\|_2^2 &= \\ &= \sum_{i=1}^d \sigma_i^2(\mathbf{X}) - \sum_{i=1}^k \sigma_i^2(\mathbf{X}) = \sum_{i=k+1}^d \sigma_i^2(\mathbf{X}) = \text{OPT}_k. \end{aligned}$$

Also, we used that $\sum_{i=1}^k \|\mathbf{v}_i^\top \mathbf{X}\|_2^2 \leq \sum_{i=1}^k \sigma_i^2(\mathbf{X}) \leq \|\mathbf{X}\|_F^2$, and that $\sum_{i=1}^k \|\mathbf{v}_i^\top \mathbf{R}\|_2^2 \leq k \|\mathbf{R}\|_2^2$. The latter inequality is true because $\|\mathbf{v}_i^\top \mathbf{R}\|_2^2 \leq \|\mathbf{R}\|_2^2$ for all $i = 1, \dots, k$. ■

Next, we provide an upper bound on the spectral norm squared of $\mathbf{R}$. The previous lemma motivates the need for such an analysis.

LEMMA 3.6. *Let $\mathbf{X}, \mathbf{R} \in \mathbb{R}^{d \times n}$, be the matrices whose t 'th column is $\mathbf{x}_t$ and $\mathbf{r}_t \in \mathbb{R}^d$ (the vectors $\mathbf{r}_t$'s are taken as the ones in the end of the corresponding iteration), respectively. Then,*

$$\|\mathbf{R}\|_2^2 \leq 2\|\mathbf{X}\|_F^2/\ell.$$

Proof. Notice that according to the updates of $\mathbf{C}$ we have that

$$\mathbf{R}\mathbf{R}^\top = \mathbf{C}_z + \sum_{j=1}^{\ell'} \lambda_j \mathbf{u}_j \mathbf{u}_j^\top.$$

According to Lemma 3.1 we have that

$$\mathbf{C}_z \cdot \left(\sum_{j=1}^{\ell'} \lambda_j \mathbf{u}_j \mathbf{u}_j^\top \right) = \mathbf{0}_{d \times d}$$

meaning that

$$\|\mathbf{R}\|_2^2 = \|\mathbf{R}\mathbf{R}^\top\| = \max \left\{ \|\mathbf{C}_z\|, \left\| \sum_{j=1}^{\ell'} \lambda_j \mathbf{u}_j \mathbf{u}_j^\top \right\| \right\}.$$

The second equality is proved in Lemma 5.4 in Section 5. Now, according to Lemma 3.2 we have that $\|\sum_{j=1}^{\ell'} \lambda_j \mathbf{u}_j \mathbf{u}_j^\top\| \leq 2\|\mathbf{X}\|_F^2/\ell$. From the structure of the algorithm it is straightforward that upon termination it must be the case that $\|\mathbf{C}_z\| \leq 2\|\mathbf{X}\|_F^2/\ell$, as $\|\mathbf{C}\| \leq \|\mathbf{C} + \mathbf{r}_t \mathbf{r}_t^\top\|$ and the if the latter is larger than $2\|\mathbf{X}\|_F^2/\ell$ then the while loop in the algorithm will not halt. Combining the three inequalities leads to the claim. ■

Finally, we provide an upper bound on the number of times the algorithm enters the while-loop; this immediately implies an upper bound on the number of vectors $\mathbf{u}$ inserted in $\mathbf{U}$. The main idea here is to study the trace of the matrix $\sum_{j=1}^{\ell'} \lambda_j \mathbf{u}_j \mathbf{u}_j^\top$ and provide a lower bound that depends on ℓ' and an upper bound that depends on ℓ . Combining the two bounds gives the desired relation between ℓ and ℓ' .

LEMMA 3.7. *Let $\mathbf{X} \in \mathbb{R}^{d \times n}$ be the matrix whose t 'th column is $\mathbf{x}_t$. Then, assuming that for all t , $\|\mathbf{x}_t\|_2^2 \leq \|\mathbf{X}\|_F^2/\ell$, the while loop in Algorithm 1 can occur a total of at most $\ell' \leq \ell \cdot \min\{1, \text{OPT}_k / \|\mathbf{X}\|_F^2 + \sqrt{8k/\ell}\} \leq \ell$.*

Proof. For notational convenience, let $\sum_{j=1}^{\ell'} \lambda_j \mathbf{u}_j \mathbf{u}_j^\top := \mathbf{Z}$. First, using Lemma 3.3, we calculate a lower bound

for $\text{Trace}(\mathbf{Z})$:

$$(3.6) \quad \text{Trace}(\mathbf{Z}) = \sum_{j=1}^{\ell'} \lambda_j \geq \ell' \cdot (\|\mathbf{X}\|_F^2/\ell).$$

Since $\mathbf{C}_z$ is clearly positive semi definite we have

$$\text{Trace}(\mathbf{Z}) = \text{Trace}(\mathbf{R}\mathbf{R}^\top) - \text{Trace}(\mathbf{C}_z) \leq$$

$$(3.7) \quad \text{Trace}(\mathbf{R}\mathbf{R}^\top) = \|\mathbf{R}\|_F^2 \leq \|\mathbf{X}\|_F^2.$$

The inequality $\|\mathbf{R}\|_F^2 \leq \|\mathbf{X}\|_F^2$ follows because for all t ,

$$\|\mathbf{r}_t\|_2^2 = \|(\mathbf{I}_d - \mathbf{U}_t \mathbf{U}_t^\top) \mathbf{x}_t\|_2^2 \leq$$

$$\|(\mathbf{I}_d - \mathbf{U}_t \mathbf{U}_t^\top)\|_2^2 \cdot \|\mathbf{x}_t\|_2^2 = \|\mathbf{x}_t\|_2^2,$$

since $\|\mathbf{I}_d - \mathbf{U}_t \mathbf{U}_t^\top\|_2^2 \leq 1$. Also, by Theorem 2.1:

$$\|\mathbf{R}\|_F^2 \leq \text{OPT}_k + \sqrt{8k/\ell} \|\mathbf{X}\|_F^2,$$

hence

$$(3.8) \quad \text{Trace}(\mathbf{Z}) \leq \|\mathbf{R}\|_F^2 \leq \text{OPT}_k + \sqrt{8k/\ell} \|\mathbf{X}\|_F^2.$$

Combining Eqn. (3.6), Eqn. (3.7), and Eqn. (3.8) shows the claim.

4 A complex and efficient OPCA algorithm

In this section we modify Algorithm 1 in order to (1) remove the assumption of knowledge of $\|\mathbf{X}\|_F^2$ and (2) improve the time and space complexity. These two improvements are made at the expense of sacrificing the accuracy *slightly*. To sidestep trivial technicalities, we assume some prior knowledge over the quantity $\|\mathbf{X}\|_F^2$; we merely assume that we have some quantity w_0 such that $w_0 \leq \|\mathbf{X}\|_F^2$ and $w_0 \gg \|\mathbf{x}_t\|^2$ for all $\|\mathbf{x}_t\|_2^2$. The assumption of knowing w_0 can be removed by working with a buffer of roughly k/ε^3 columns. Since we already require this amount of space, the resulting increase in the memory complexity is asymptotically negligible.

Specifically, we provide the following theorem:

THEOREM 4.1. *There exists an algorithm that on input $\mathbf{X} \in \mathbb{R}^{d \times n}$, target dimension $k < d$, and $\varepsilon \in (0, 1/15)$, it produces vectors $\mathbf{y}_1, \mathbf{y}_2, \dots, \mathbf{y}_n \in \mathbb{R}^\ell$, with $\min_{\Phi \in \mathcal{O}_{d,\ell}} \sum_{t=1}^n \|\mathbf{x}_t - \Phi \mathbf{y}_t\|_2^2 \leq \text{OPT}_k + \varepsilon \|\mathbf{X}\|_F^2$. For some $\delta = \min\{\varepsilon, \varepsilon^2 \|\mathbf{X}\|_F^2 / \text{OPT}\}$, this algorithm requires $O(dk/\delta^3)$ space and $O(ndk/\delta^3 + \log(n)dk^3/\delta^6)$ arithmetic operations assuming $\|\mathbf{x}_t\|_2$ are polynomial in n . The target dimension of the algorithm is $\ell = k/\delta^3$.*

Algorithm 2 An efficient online PCA algorithm

input: $\mathbf{X}$, k , $\varepsilon \in (0, \frac{1}{15})$, w_0 (which defaults to $w_0 = 0$)
 $\mathbf{U} = \mathbf{0}^{d \times k/\varepsilon^3}$, $\mathbf{Z} = \mathbf{0}^{d \times k/\varepsilon^2}$, $w = 0$, $w_{\mathbf{U}} = \mathbf{0}^{k/\varepsilon^3}$
for $t = 1, \dots, n$ **do**
 $w = w + \|\mathbf{x}_t\|_2^2$
 $\mathbf{r}_t = \mathbf{x}_t - \mathbf{U}\mathbf{U}^\top \mathbf{x}_t$
 $\mathbf{C} = (\mathbf{I} - \mathbf{U}\mathbf{U}^\top)\mathbf{Z}\mathbf{Z}^\top(\mathbf{I} - \mathbf{U}\mathbf{U}^\top)$
 while $\|\mathbf{C} + \mathbf{r}_t \mathbf{r}_t^\top\|_2 \geq \max\{w_0, w\} \cdot \frac{\varepsilon^2}{k}$ **do**
 $\mathbf{u}, \lambda = \text{topEigenvectorAndValue}(\mathbf{C})$
 $(w_{\mathbf{U}})_{\mathbf{u}} \leftarrow \lambda$
 If $\mathbf{U}$ has a zero column, write $\mathbf{u}$ in its place.
 Otherwise, write $\mathbf{u}$ instead of the column $\mathbf{v}$ of $\mathbf{U}$ with the minimal quantity of $(w_{\mathbf{U}})_{\mathbf{v}}$
 $\mathbf{C} = (\mathbf{I} - \mathbf{U}\mathbf{U}^\top)\mathbf{Z}\mathbf{Z}^\top(\mathbf{I} - \mathbf{U}\mathbf{U}^\top)$
 $\mathbf{r}_t = \mathbf{x}_t - \mathbf{U}\mathbf{U}^\top \mathbf{x}_t$
 end while
 $\mathbf{Z} \leftarrow \text{Sketch}([\mathbf{Z}, \mathbf{r}_t])$
 For each non-zero column $\mathbf{u}$ in $\mathbf{U}$, $(w_{\mathbf{U}})_{\mathbf{u}} \leftarrow (w_{\mathbf{U}})_{\mathbf{u}} + \langle \mathbf{x}_t, \mathbf{u} \rangle^2$
 yield: $\mathbf{y}_t \leftarrow \mathbf{U}^\top \mathbf{x}_t$
end for

4.1 Notations For a variable ξ in the algorithm, where ξ can either be a matrix $\mathbf{Z}, \mathbf{X}, \mathbf{C}$ or a vector $\mathbf{r}, \mathbf{x}$ we denote by ξ_t the value of ξ at the end of iteration number t . ξ_0 will denote the value of the variable at the beginning of the algorithm. For variables that may change during the while loop we denote by $\xi_{t,z}$ the value of ξ at the end of the z 'th while loop in the t 'th iteration. In particular, if the while loop was never entered the value of ξ at the end of the iteration t will be equal to $\xi_{t,0}$. For such ξ that may change during the while loop notice that ξ_t is its value at the end of the iteration, meaning after the last while loop has ended.

An exception to the above is for the variable w . Here, we denote by w_t the value of $\max\{w_0, w\}$ at the end of iteration number t . We denote by n the index of the last iteration of the algorithm. In particular ξ_n denotes the value of ξ upon the termination of the algorithm.

4.2 Matrix Sketching We use the Frequent Direction matrix sketching algorithm presented by Liberty in [13]. This algorithm provides a sketching algorithm for the streaming version where we observe a matrix $\mathbf{R}$ column by column. The idea is to maintain a matrix $\mathbf{Z}$ of low rank that at all times has the property that $\mathbf{Z}\mathbf{Z}^\top$ and $\mathbf{R}\mathbf{R}^\top$ are close, in operator norm.

LEMMA 4.1. Let $\mathbf{R}_1, \dots, \mathbf{R}_t, \dots$ be a sequence of matrices with columns of dimension d where $\mathbf{R}_{t+1}$ is ob-

tained by appending a column vector $\mathbf{r}_{t+1}$ to $\mathbf{R}_t$. Let $\mathbf{Z}_1, \dots, \mathbf{Z}_t, \dots$ the corresponding sketches of $\mathbf{R}$ obtained by adding these columns as they arrive, according to the Frequent Direction algorithm in [13] with parameter ℓ .

1. The worst case time required by the sketch to add a single column is $O(\ell d)$.
2. Each $\mathbf{Z}_t$ is a matrix of dimensions $d \times O(\ell)$.
3. Let $\mathbf{u}$ be a left singular vector of $\mathbf{Z}_t$ with singular value σ and assume that $\mathbf{r}_{t+1}$ is orthogonal to $\mathbf{u}$. Then $\mathbf{u}$ is a singular vector of $\mathbf{Z}_{t+1}$ with singular value $\leq \sigma$.
4. For any vector $\mathbf{u}$ and time t it holds that $\|\mathbf{Z}_t \mathbf{u}\| \leq \|\mathbf{R}_t \mathbf{u}\|$.
5. For any t there exists a positive semidefinite matrix $\mathbf{E}_t$ such that $\|\mathbf{E}_t\|_2^2 \leq \|\mathbf{R}_t\|_F^2 / \ell$ and $\mathbf{Z}_t \mathbf{Z}_t^\top + \mathbf{E} = \mathbf{R}_t \mathbf{R}_t^\top$.

4.3 Proof of Theorem 4.1

LEMMA 4.2. At all times, the matrix $\mathbf{U}^\top \mathbf{U}$ is a diagonal matrix with either zeros or ones across the diagonal. In other words, the non-zero column vectors of $\mathbf{U}$, at any time point are orthonormal.

Proof. We prove the claim for any $\mathbf{U}_{t,z}$ by induction on (t, z) . For $t = 0$ the claim is trivial as $\mathbf{U}_0 = \mathbf{0}$. For the induction step we need only to consider $\mathbf{U}_{t,z}$ for $z > 0$ since $\mathbf{U}_{t,0} = \mathbf{U}_{t-1}$. Let $\mathbf{u}$ be the new vector in $\mathbf{U}_{t,z}$. Since $\mathbf{u}$ is defined as an eigenvector of

$$\mathbf{C}_{t,z-1} = (\mathbf{I} - \mathbf{U}_{t,z-1} \mathbf{U}_{t,z-1}^\top) \mathbf{Z}_{t-1} \mathbf{Z}_{t-1}^\top (\mathbf{I} - \mathbf{U}_{t,z-1} \mathbf{U}_{t,z-1}^\top),$$

we have that $\mathbf{U}_{t,z-1} \mathbf{u} = \mathbf{0}$ and the claim immediately follows.

LEMMA 4.3. For all t, z , the non-zero columns of $\mathbf{U}_{t,z}$ are left singular vectors of $\mathbf{Z}_{t-1}$ (possibly with singular value zero). In particular, the claim holds for the final values of the matrices $\mathbf{U}$ and $\mathbf{Z}$.

Proof. We prove the claim by induction on t, z , with a trivial base case of $t = 1$ where $\mathbf{Z}_{t-1} = \mathbf{0}$. Let $t > 1$. For $z = 0$, each non-zero column vector $\mathbf{u}$ of $\mathbf{U}_{t,0}$ is a non-zero column vector of $\mathbf{U}_{t-1,z}$ for the largest valid z w.r.t $t - 1$. By the induction hypothesis it holds that $\mathbf{u}$ is a singular vector of $\mathbf{Z}_{t-2}$. According to Observation 4.2 we have that $\mathbf{r}_{t-1}$, the vector added to $\mathbf{Z}_{t-2}$ is orthogonal to $\mathbf{u}$, hence Lemma 4.1, item 3 indicates that $\mathbf{u}$ is a singular vector of $\mathbf{Z}_{t-1}$ as required.

Consider now $z > 0$. In this case $\mathbf{u}$ is a vector added in the while loop. Recall that $\mathbf{u}$ is defined as an eigenvector of

$$\mathbf{C}_{t,z-1} = (\mathbf{I} - \mathbf{U}_{t,z-1} \mathbf{U}_{t,z-1}^\top) \mathbf{Z}_{t-1} \mathbf{Z}_{t-1}^\top (\mathbf{I} - \mathbf{U}_{t,z-1} \mathbf{U}_{t,z-1}^\top)$$

According to our induction hypothesis, all of the non-zero column vectors of $\mathbf{U}_{t,z-1}$ are singular vectors of $\mathbf{Z}_{t-1}$, hence

$$\begin{aligned} \mathbf{Z}_{t-1} \mathbf{Z}_{t-1}^\top &= \\ \mathbf{C}_{t,z-1} + \mathbf{U}_{t,z-1} \mathbf{U}_{t,z-1}^\top \mathbf{Z}_{t-1} \mathbf{Z}_{t-1}^\top \mathbf{U}_{t,z-1} \mathbf{U}_{t,z-1}^\top. \end{aligned}$$

The two equalities above imply that any eigenvector of $\mathbf{C}_{t,z-1}$ is an eigenvector of $\mathbf{Z}_{t-1} \mathbf{Z}_{t-1}^\top$ as well. It follows that $\mathbf{u}$ is a singular vector of $\mathbf{Z}_{t-1}$ as required, thus proving the claim. ■

LEMMA 4.4. Let $\mathbf{v}$ be a column vector of $\mathbf{U}_{t,z}$ that is not in $\mathbf{U}_{t,z+1}$. Let $(t_{\mathbf{v}}, \mathbf{Z}_{\mathbf{v}})$ be the earliest time stamp from which $\mathbf{v}$ was a column vector in $\mathbf{U}$ consecutively up to time (t, z) . It holds that

$$\|\mathbf{Z}_{t-1} \mathbf{v}\|_2^2 + \|(\mathbf{X}_{t-1} - \mathbf{X}_{t_{\mathbf{v}}-1}) \mathbf{v}\|_2^2 \leq \frac{2\varepsilon^3}{k} w_{t-1}$$

Proof. We denote by $(w_{\mathbf{U}})_{\mathbf{u}}$ the values of the $w_{\mathbf{U}}$ vector for the different directions $\mathbf{u}$ at time (t, z) . Let $\lambda_{\mathbf{v}}$ be the eigenvalue associated with $\mathbf{v}$ during the time it was entered to $\mathbf{U}$. Then at that time $\|\mathbf{Z} \mathbf{v}\|_2^2 = \|\mathbf{C} \mathbf{v}\|_2^2 = \lambda_{\mathbf{v}}$ (recall that $\mathbf{v}$ is chosen as a vector orthogonal to those of $\mathbf{U}$ hence $\|\mathbf{C} \mathbf{v}\| = \|\mathbf{Z}(\mathbf{I} - \mathbf{U}) \mathbf{v}\|^2 = \|\mathbf{Z} \mathbf{v}\|^2$). Furthermore, since $\mathbf{v}$ was a column in $\mathbf{U}$ up to time t we get that all vectors added to $\mathbf{R}$ from the insertion of $\mathbf{v}$ up to time t are orthogonal to $\mathbf{v}$. Lemma 4.1, item 3 shows that

$$\|\mathbf{Z}_{t-1} \mathbf{v}\|_2^2 \leq \lambda_{\mathbf{v}}.$$

Since $\mathbf{v}$ was a column vector of $\mathbf{U}$ from iteration $t_{\mathbf{v}}$ up to iteration t , we have that

$$\begin{aligned} \|\mathbf{Z}_{t-1} \mathbf{v}\|_2^2 + \|(\mathbf{X}_{t-1} - \mathbf{X}_{t_{\mathbf{v}}-1}) \mathbf{v}\|_2^2 &\leq \\ \lambda_{\mathbf{v}} + \sum_{\tau=t_{\mathbf{v}}}^t \langle \mathbf{v}, \mathbf{x}_{\tau} \rangle^2 &= (w_{\mathbf{U}})_{\mathbf{v}} \end{aligned}$$

It remains to bound the quantity of $(w_{\mathbf{U}})_{\mathbf{v}}$. We will bound the sum $\sum_{\mathbf{u} \in \mathbf{U}_{t,z}} (w_{\mathbf{U}})_{\mathbf{u}}$ and use the fact that $\mathbf{v}$ is the minimizer of the corresponding expression hence $(w_{\mathbf{U}})_{\mathbf{v}}$ is upper bounded by $\sum_{\mathbf{u}} (w_{\mathbf{U}})_{\mathbf{u}} \cdot \frac{\varepsilon^3}{k}$.

Let $t_{\mathbf{u}}$ be the index of the iteration in which $\mathbf{u}$ is inserted into $\mathbf{U}$. It is easy to verify that for $\lambda_{\mathbf{u}} = \|\mathbf{C}_{t_{\mathbf{u}}-1} \mathbf{u}\|_2^2$ it holds that

$$w_{\mathbf{u}} = \lambda_{\mathbf{u}} + \|(\mathbf{X}_{t-1} - \mathbf{X}_{t_{\mathbf{u}}-1}) \mathbf{u}\|_2^2$$

Now, since $\mathbf{C} \preceq \mathbf{Z} \preceq \mathbf{R}$ we have that

$$w_{\mathbf{u}} = \|\mathbf{R}_{t_{\mathbf{u}}-1} \mathbf{u}\|_2^2 + \|(\mathbf{X}_{t-1} - \mathbf{X}_{t_{\mathbf{u}}-1}) \mathbf{u}\|_2^2$$

Finally,

$$\sum_{\mathbf{u}} (w_{\mathbf{U}})_{\mathbf{u}} \leq \sum_{\mathbf{u}} \|\mathbf{R}_{t_{\mathbf{u}}-1} \mathbf{u}\|_2^2 + \sum_{\mathbf{u}} \|(\mathbf{X}_{t-1} - \mathbf{X}_{t_{\mathbf{u}}-1}) \mathbf{u}\|_2^2 \stackrel{(i)}{\leq}$$

$$\sum_{\mathbf{u}} \|\mathbf{X}_{t-1} \mathbf{u}\|_2^2 + \|\mathbf{R}_{t-1} \mathbf{u}\|_2^2 \stackrel{(ii)}{\leq} \|\mathbf{X}_{t-1}\|_F^2 + \|\mathbf{R}_{t-1}\|_F^2$$

$$\stackrel{(iii)}{\leq} 2\|\mathbf{X}_{t-1}\|_F^2 = 2w_{t-1}$$

Inequality (i) is immediate from the definitions of $\mathbf{X}_t, \mathbf{R}_t$ as concatenations of t column vectors. Inequality (ii) follows since any orthogonal vector set and any matrix admit $\sum_{\mathbf{u}} \|\mathbf{A} \mathbf{u}\|_2^2 \leq \|\mathbf{A}\|_F^2$. Inequality (iii) is due to the fact that each column of $\mathbf{R}$ is obtained by projection a column of $\mathbf{X}$ onto some subspace. ■

LEMMA 4.5. At all times $\|\mathbf{C}_{t,z}\|_2 \leq w_{t-1} \cdot \frac{\varepsilon^2}{k}$,

Proof. We prove the claim by induction over t, z . The base case for $t = 0$ is trivial. For $t > 0, z = 0$, if the while loop in iteration t was entered to we have that $\mathbf{C}_{t,0} = \mathbf{C}_{t-1,z}$ for some z . Since $w_{t-1} \geq w_{t-2}$ the claim holds. If t is such that the while loop of the iteration was not entered the condition of the while loop asserts that $\|\mathbf{C}_{t,z}\| = \|\mathbf{C}_t\| \leq w_{t-1} \cdot \frac{\varepsilon^2}{k}$.

Consider now $t, z > 0$. We have that

$$\mathbf{C}_{t,z-1} = (\mathbf{I}_d - \mathbf{U}_{t,z-1} \mathbf{U}_{t,z-1}^\top) \mathbf{Z}_{t-1} \mathbf{Z}_{t-1}^\top (\mathbf{I}_d - \mathbf{U}_{t,z-1} \mathbf{U}_{t,z-1}^\top)$$

$$\mathbf{C}_{t,z} = (\mathbf{I}_d - \mathbf{U}_{t,z} \mathbf{U}_{t,z}^\top) \mathbf{Z}_{t-1} \mathbf{Z}_{t-1}^\top (\mathbf{I}_d - \mathbf{U}_{t,z} \mathbf{U}_{t,z}^\top)$$

If $\mathbf{U}_{t,z}$ is obtained by writing $\mathbf{u}$ instead of a zero column of $\mathbf{U}_{t,z-1}$ then $\mathbf{C}_{t,z}$ is a projection of $\mathbf{C}_{t,z-1}$ and the claim holds due to the induction hypothesis. If not, $\mathbf{u}$ is inserted instead of some vector $\mathbf{v}$. According to Lemmas 4.3 and 4.4, $\mathbf{v}$ is an eigenvector of $\mathbf{Z}_{t-1} \mathbf{Z}_{t-1}^\top$ with eigenvalue $\lambda_{\mathbf{v}} \leq w_{t-1} \cdot \frac{2\varepsilon^3}{k} \leq w_{t-1} \cdot \frac{\varepsilon^2}{k}$, assuming $\varepsilon \leq 0.5$. It follows that $\mathbf{C}_{t,z}$ is a projection of $\mathbf{C}_{t,z-1} + \lambda_{\mathbf{v}} \mathbf{v} \mathbf{v}^\top$. Now, since $\mathbf{C}_{t,z-1} \mathbf{v} = 0$ (as $\mathbf{v}$ is a column vector of $\mathbf{U}_{t,z-1}$) we have that

$$\|\mathbf{C}_{t,z}\|_2 \leq \|\mathbf{C}_{t,z-1} + \lambda_{\mathbf{v}} \mathbf{v} \mathbf{v}^\top\|_2 = \max\{\|\mathbf{C}_{t,z-1}\|_2, \|\lambda_{\mathbf{v}} \mathbf{v} \mathbf{v}^\top\|_2\}$$

According to our induction hypothesis and the bound for $\lambda_{\mathbf{v}}$, the above expression is bounded by $w_{t-1} \cdot \frac{\varepsilon^2}{k}$ as required. ■

LEMMA 4.6. Let $\mathbf{u}$ be a vector that is not in $\mathbf{U}_{t,z}$ and in $\mathbf{U}_{t,z+1}$. Let λ be the eigenvector associated to it w.r.t $\mathbf{C}_{t,z}$. It holds that $\lambda \leq w_{t-1} \cdot \frac{\varepsilon^2}{k}$. Furthermore, if $\mathbf{u}$ is a column vector in $\mathbf{U}_{t',z'}$ for all $t \leq t', z \leq z'$, it holds that $\|\mathbf{u}^\top \mathbf{Z}_n\|^2 \leq \lambda \leq \|\mathbf{X}_n\|_F^2 \cdot \frac{\varepsilon^2}{k}$.

Proof. Since $\mathbf{u}$ is chosen as the top eigenvector of $\mathbf{C}_{t,z}$ we have by Lemma 4.5 that

$$\lambda = \|\mathbf{C}_{t,z} \mathbf{u}\|_2 = \|\mathbf{C}_{t,z}\|_2 \leq w_{t-1} \cdot \frac{\varepsilon^2}{k}$$

For the second claim in the lemma we note that since $\mathbf{u}$ is an eigenvector of $\mathbf{C}_{t,z-1} = (\mathbf{I} - \mathbf{U}_{t,z-1}\mathbf{U}_{t,z-1}^\top)\mathbf{Z}_{t-1}\mathbf{Z}_{t-1}^\top(\mathbf{I} - \mathbf{U}_{t,z-1}\mathbf{U}_{t,z-1}^\top)$, we have that $\mathbf{U}_{t,z-1}^\top \mathbf{u} = 0$, hence

$$\|\mathbf{u}^\top \mathbf{Z}_{t-1}\|_2^2 = \|\mathbf{u}^\top (\mathbf{I} - \mathbf{U}_{t,z-1}\mathbf{U}_{t,z-1}^\top)\mathbf{Z}_{t-1}\|_2^2 =$$

$$\mathbf{u}^\top \mathbf{C}_{t,z} \mathbf{u} \leq \|\mathbf{C}_{t,z} \mathbf{u}\|_2 \leq w_{t-1} \cdot \frac{\varepsilon^2}{k}$$

Since $\mathbf{u}$ is assumed to be an element of $\mathbf{U}$ throughout the running time of the algorithm, it holds that for all future vectors $\mathbf{r}$ added to the sketch $\mathbf{Z}$, $\mathbf{u}$ is orthogonal to $\mathbf{r}$. The claim now follows from Lemma 4.1 item 3. ■

LEMMA 4.7. $\|\mathbf{R}_n\|_2^2 \leq \frac{2\varepsilon^2}{k} \|\mathbf{X}_n\|_F^2$.

Proof. Let $\mathbf{u}_1, \dots, \mathbf{u}_\ell$ and $\lambda_1, \dots, \lambda_\ell$ be the columns of $\mathbf{U}_n$ and their corresponding eigenvalues in $\mathbf{C}$ at the time of their addition to $\mathbf{U}$. From Lemmas 4.3 and 4.6 we have that each $\mathbf{u}_j$ is an eigenvector of $\mathbf{Z}\mathbf{Z}^\top$ with eigenvalue $\lambda'_j \leq \lambda_j \leq \frac{\varepsilon^2}{k} \|\mathbf{X}_n\|_F^2$. It follows that

$$\|\mathbf{Z}_n \mathbf{Z}_n^\top\|_2 =$$

$$\begin{aligned} & \max \left\{ \frac{\varepsilon^2}{k} \|\mathbf{X}_n\|_F^2, \|(\mathbf{I} - \mathbf{U}_n \mathbf{U}_n^\top) \mathbf{Z}_n \mathbf{Z}_n^\top (\mathbf{I} - \mathbf{U}_n \mathbf{U}_n^\top)\|_2 \right\} \\ & = \max \left\{ \frac{\varepsilon^2}{k} \|\mathbf{X}_n\|_F^2, \|\mathbf{C}_n\|_2 \right\} \leq \frac{\varepsilon^2}{k} \|\mathbf{X}_n\|_F^2. \end{aligned}$$

The last inequality is due to Lemma 4.5.

Next, by the sketching property (Lemma 4.1 item 5), for appropriate matrix $\mathbf{E}$: $\mathbf{Z}_n \mathbf{Z}_n^\top = \mathbf{R}_n \mathbf{R}_n^\top + \mathbf{E}$, with $\|\mathbf{E}\| \leq \frac{\varepsilon^2}{k} \|\mathbf{R}_n\|_F^2$. As the columns of $\mathbf{R}$ are projections of those of $\mathbf{X}$ we have that $\|\mathbf{R}_n\|_F^2 \leq \|\mathbf{X}_n\|_F^2$, hence

$$\|\mathbf{R}_n\|_2^2 = \|\mathbf{R}_n \mathbf{R}_n^\top\|_2 =$$

$$\|\mathbf{Z}_n \mathbf{Z}_n^\top - \mathbf{E}\|_2 \leq \|\mathbf{Z}_n \mathbf{Z}_n^\top\|_2 + \|\mathbf{E}\|_2 \leq \frac{2\varepsilon^2}{k} \|\mathbf{X}_n\|_F^2,$$

which concludes the proof. ■

LEMMA 4.8.

$$\|\mathbf{R}_n\|_F^2 \leq \text{OPT}_k + 4\varepsilon \|\mathbf{X}_n\|_F^2$$

Proof. The Lemma can be proven analogically to Theorem 2.1 as the only difference is the bound over $\|\mathbf{R}_n\|_2^2$. ■

LEMMA 4.9. Assume that for all t , $\|\mathbf{x}_t\|_2^2 \leq w_t \cdot \frac{\varepsilon^2}{5k}$. Assume that $\varepsilon \leq 0.1$. For $\tau > 0$ consider the iterations of the algorithm during which $w_t \in [2^\tau, 2^{\tau+1})$. During this time, the while loop will be executed at most $5k/\varepsilon^2$ times.

Proof. For the proof, we define a potential function

$$\Phi_{t,z} = \text{Trace}(\mathbf{R}_{t-1} \mathbf{R}_{t-1}^\top) - \text{Trace}(\mathbf{C}_{t,z}).$$

We first notice that since $\mathbf{C}$ is clearly PSD,

$$\Phi_{t,z} \leq \text{Trace}(\mathbf{R}_{t-1} \mathbf{R}_{t-1}^\top) = \|\mathbf{R}_{t-1}\|_F^2 \leq \|\mathbf{X}_{t-1}\|_F^2 \leq 2^{\tau+1}.$$

The first inequality correct because the columns of $\mathbf{R}$ are projections of those of $\mathbf{X}$ and the second because $\|\mathbf{X}_{t-1}\|_F^2 \leq w_{t-1}$. We show that Φ is non-decreasing with time and that, for valid $z > 0$, $\Phi_{t,z} \geq \Phi_{t,z-1} + 0.2 \frac{\varepsilon^2}{k} 2^{\tau+1}$. The result immediately follows.

Consider a pair (t, z) followed by the pair of indices $(t+1, 0)$. Here, $\Phi_{t+1,0} - \Phi_{t,z} = \|\mathbf{r}_t\|_2^2 \geq 0$ hence for such pairs the potential is non-decreasing. Now consider some pair (t, z) for $z > 0$. Since (t, z) is a valid pair it holds that

$$\begin{aligned} & \|\mathbf{C}_{t,z-1}\|_2 \\ & \geq \|\mathbf{C}_{t,z-1} + \mathbf{r}_{t,z-1} \mathbf{r}_{t,z-1}^\top - \mathbf{r}_{t,z-1} \mathbf{r}_{t,z-1}^\top\|_2 \\ & \geq \|\mathbf{C}_{t,z-1} + \mathbf{r}_{t,z-1} \mathbf{r}_{t,z-1}^\top\|_2 - \|\mathbf{r}_{t,z-1} \mathbf{r}_{t,z-1}^\top\|_2 \\ & \geq w_t \frac{\varepsilon^2}{k} (1 - 0.2) \\ (4.9) \quad & \geq 0.4 \cdot 2^{\tau+1} \frac{\varepsilon^2}{k} \end{aligned}$$

Denote by $\mathbf{u}$ the column vector in $\mathbf{U}_{t,z}$ that is not in $\mathbf{U}_{t,z-1}$. Let $\mathbf{U}'$ be the matrix obtained by appending the column $\mathbf{u}$ to the matrix $\mathbf{U}_{t,z-1}$. Let

$$\begin{aligned} \mathbf{C}' &= (\mathbf{I} - \mathbf{U}'(\mathbf{U}')^\top) \mathbf{Z}_{t-1} \mathbf{Z}_{t-1}^\top (\mathbf{I} - \mathbf{U}'(\mathbf{U}')^\top) = \\ & (\mathbf{I} - \mathbf{u} \mathbf{u}^\top) \mathbf{C}_{t,z-1} (\mathbf{I} - \mathbf{u} \mathbf{u}^\top) \end{aligned}$$

Since $\mathbf{u}$ is the top eigenvector of $\mathbf{C}_{t,z-1}$ we have by equation (4.9) that

$$\text{Trace}(\mathbf{C}') - \text{Trace}(\mathbf{C}_{t,z-1}) = \|\mathbf{C}_{t,z-1}\|_2 \geq 0.4 \cdot 2^{\tau+1} \frac{\varepsilon^2}{k}$$

If $\mathbf{U}_{t,z-1}$ had a zero column then $\mathbf{C}_{t,z} = \mathbf{C}'$ and we are done. If not, let $\mathbf{v}$ be the vector that was replaced by $\mathbf{u}$. According to Lemma 4.3, $\mathbf{v}$ is a singular vector of $\mathbf{Z}_{t-1}$. According to Lemma 4.4 and $\varepsilon < 0.1$ we have that

$$\|\mathbf{Z}_{t-1} \mathbf{v}\|_2^2 \leq \frac{2\varepsilon^3}{k} \|\mathbf{X}_{t-1}\|_F^2 \leq \frac{\varepsilon^2}{5k} 2^{\tau+1}.$$

Hence,

$$\mathbf{C}_{t,z} = \mathbf{C}' + \|\mathbf{Z}_{t-1} \mathbf{v}\|^2 \cdot \mathbf{v} \mathbf{v}^\top$$

meaning that

$$\text{Trace}(\mathbf{C}_{t,z} - \mathbf{C}_{t,z-1}) =$$

$$\text{Trace}(\mathbf{C}_{t,z} - \mathbf{C}') + \text{Trace}(\mathbf{C}' - \mathbf{C}_{t,z-1}) \geq \frac{\varepsilon^2}{5k} 2^{\tau+1}.$$

We conclude that as required Φ is non-decreasing over time and in each iteration of the while loop, increases by at least $\frac{\varepsilon^2}{5k} 2^{\tau+1}$. Since Φ is upper bounded by $2^{\tau+1}$ during the discussed iterations, the lemma follows. ■

LEMMA 4.10. *Let $(\mathbf{v}_1, t'_1 + 1, t_1 + 1), \dots, (\mathbf{v}_j, t'_j + 1, t_j + 1), \dots$ be the sequence of triplets of vectors removed from $\mathbf{U}$, the times on which they were added to $\mathbf{U}$ and the times on which they were removed from $\mathbf{U}$.*

$$\text{ALG} \leq \left(\|\mathbf{R}_n\|_F + 2 \sqrt{\sum_j \|(\mathbf{X}_{t_j} - \mathbf{X}_{t'_j}) \mathbf{v}_j\|_2^2} \right)^2.$$

Proof. For any time t denote by $\mathbf{U}_t$ the matrix $\mathbf{U}$ in the end of iteration t , by $\mathbf{U}_t^{(1)}$ the outcome of zeroing-out every column of $\mathbf{U}_t$ that is different from the corresponding column in $\mathbf{U}_n$ and by $\mathbf{U}_t^{(2)}$ its complement, that is the outcome of zeroing-out every column in $\mathbf{U}_t$ that is identical to the corresponding column in $\mathbf{U}_n$. In the same way define $\mathbf{U}^{(2)}$ to be the outcome of zeroing-out columns in $\mathbf{U}_n$ that are all zeros in $\mathbf{U}_t^{(2)}$.

It holds that,

$$\begin{aligned} & \|\mathbf{x}_t - \mathbf{U}_n \mathbf{U}_t^\top \mathbf{x}_t\|_2^2 \\ &= \|\mathbf{x}_t - (\mathbf{U}_t + \mathbf{U}_n - \mathbf{U}_t) \mathbf{U}_t^\top \mathbf{x}_t\|_2^2 \\ &\leq \left(\|\mathbf{x}_t - \mathbf{U}_t \mathbf{U}_t^\top \mathbf{x}_t\|_2 + \|(\mathbf{U}_n - \mathbf{U}_t) \mathbf{U}_t^\top \mathbf{x}_t\|_2 \right)^2 \\ &= \left(\|\mathbf{r}_t\|_2 + \|(\mathbf{U}^{(2)} - \mathbf{U}_t^{(2)}) (\mathbf{U}_t^{(2)})^\top \mathbf{x}_t\|_2 \right)^2 \\ &\leq \left(\|\mathbf{r}_t\|_2 + 2 \|(\mathbf{U}_t^{(2)})^\top \mathbf{x}_t\|_2 \right)^2 \end{aligned}$$

Summing over all times t we have,

$$\begin{aligned} \text{ALG} &= \sum_{t=1}^n \left(\|\mathbf{r}_t\|_2 + 2 \|(\mathbf{U}_t^{(2)})^\top \mathbf{x}_t\|_2 \right)^2 \\ &= \sum_{t=1}^n \|\mathbf{r}_t\|_2^2 + 4 \|\mathbf{r}_t\|_2 \|(\mathbf{U}_t^{(2)})^\top \mathbf{x}_t\|_2 \\ &\quad + 4 \|(\mathbf{U}_t^{(2)})^\top \mathbf{x}_t\|_2^2 \\ &\leq \|\mathbf{R}_n\|_F^2 + 4 \sqrt{\sum_{t=1}^n \|\mathbf{r}_t\|_2^2} \sqrt{\sum_{t=1}^n \|(\mathbf{U}_t^{(2)})^\top \mathbf{x}_t\|_2^2} \\ &\quad + 4 \sum_t \|(\mathbf{U}_t^{(2)})^\top \mathbf{x}_t\|_2^2 \end{aligned}$$

Where the last inequality follows from applying the Cauchy-Schwarz inequality to the dot product between the vectors $(\|\mathbf{r}_1\|_2, \|\mathbf{r}_2\|_2, \dots, \|\mathbf{r}_n\|_2)$ and $(\|(\mathbf{U}_1^{(2)})^\top \mathbf{x}_1\|_2, \|(\mathbf{U}_2^{(2)})^\top \mathbf{x}_2\|_2, \dots, \|(\mathbf{U}_n^{(2)})^\top \mathbf{x}_n\|_2)$.

Since $\mathbf{U}_t^{(2)}$ contains only vectors that were columns of $\mathbf{U}$ at time t but were replaced later, and are not present in $\mathbf{U}_n$, we have that

$$\|(\mathbf{U}_t^{(2)})^\top \mathbf{x}_t\|_2^2 \leq \sum_{j:t_j > t > t'_j} (\mathbf{x}_t^\top \mathbf{v}_j)^2$$

and so

$$\sum_{t=1}^n \|(\mathbf{U}_t^{(2)})^\top \mathbf{x}_t\|_2^2 \leq \sum_j \|(\mathbf{X}_{t_j} - \mathbf{X}_{t'_j}) \mathbf{v}_j\|_2^2.$$

Thus we have that, $\text{ALG} \leq$

$$\begin{aligned} &\leq \|\mathbf{R}_n\|_F^2 + 4 \|\mathbf{R}_n\|_F \sqrt{\sum_j \|(\mathbf{X}_{t_j} - \mathbf{X}_{t'_j}) \mathbf{v}_j\|_2^2} \\ &\quad + 4 \sum_j \|(\mathbf{X}_{t_j} - \mathbf{X}_{t'_j}) \mathbf{v}_j\|_2^2 \\ &= \left(\|\mathbf{R}_n\|_F + 2 \sqrt{\sum_j \|(\mathbf{X}_{t_j} - \mathbf{X}_{t'_j}) \mathbf{v}_j\|_2^2} \right)^2, \end{aligned}$$

which concludes the proof. ■

LEMMA 4.11. *Let $(\mathbf{v}_1, t'_1 + 1, t_1 + 1), \dots, (\mathbf{v}_j, t'_j + 1, t_j + 1), \dots$ be the sequence of triplets of vectors removed from $\mathbf{U}$, the times on which they were added to $\mathbf{U}$ and the times on which they were removed from $\mathbf{U}$. Then*

$$\sum_j \|(\mathbf{X}_{t_j} - \mathbf{X}_{t'_j}) \mathbf{v}_j\|_2^2 \leq 20\varepsilon \|\mathbf{X}_n\|_F^2.$$

Proof. For some $\tau > 0$ consider the execution of the algorithm during the period in which $w_t \in [2^\tau, 2^{\tau+1})$. According to Lemma 4.9, at most $\frac{5k}{\varepsilon^2}$ vectors $\mathbf{v}$ were removed from the $\mathbf{U}$ during that period. According to Lemma 4.4, for each such $\mathbf{v}_j$ it holds that

$$\|(\mathbf{X}_{t_j} - \mathbf{X}_{t'_j}) \mathbf{v}_j\|_2^2 \leq w_{t_j} \cdot \frac{2\varepsilon^3}{k} \leq \frac{2\varepsilon^3}{k} 2^{\tau+1}$$

It follows that the contribution of vectors $\mathbf{v}$ thrown from the set during the discussed time period is at most

$$5 \frac{k}{\varepsilon^2} \cdot \frac{2\varepsilon^3}{k} 2^{\tau+1} = 10\varepsilon \cdot 2^{\tau+1}$$

The entire sum can now be bounded by a geometric series, ending at $\tau = \log_2(\|\mathbf{X}\|_F^2)$ thus proving the lemma. ■

COROLLARY 4.1.

$$\text{ALG}_{k,\varepsilon} \leq \left(\sqrt{\text{OPT}_k} + \left(\sqrt{4} + \sqrt{20} \right) \sqrt{\varepsilon} \|\mathbf{X}_n\|_F \right)^2$$

$$\leq \left(\sqrt{\text{OPT}_k} + 6.48\sqrt{\varepsilon} \|\mathbf{X}_n\|_F \right)^2$$

In particular, for $\varepsilon = \delta^2 \cdot \text{OPT}_k / \|\mathbf{X}_n\|_F^2$,

$$\text{ALG}_{k,\varepsilon} = \text{OPT}_k(1 + O(\delta))$$

LEMMA 4.12. (TIME COMPLEXITY) *Algorithm 2 requires $O(n \frac{dk}{\varepsilon^3} + \log(w_n/w_0) \frac{dk^3}{\varepsilon^6})$ arithmetic operations.*

Proof. We begin by pointing out that in the algorithm we work with the $d \times d$ matrices $\mathbf{C}$ and $\mathbf{ZZ}^\top$. These matrices have a bounded rank and furthermore, we are always able to maintain a representation of them of the form $\mathbf{AA}^\top$ with $\mathbf{A}$ being a $d \times r$ matrix with $\mathbf{R}$ being the rank of $\mathbf{C}$ or $\mathbf{ZZ}^\top$. Hence, all computations involving them requires $O(dr)$ or $O(dr^2)$ arithmetic operations (corresponding to a product with a vector and computing the eigendecomposition).

Consider first the cost of the operations outside the while loop that do not involve the matrix $\mathbf{C}$. It is easy to see, with Lemma 4.1, item 1, that the amortized time required in each iteration is $O(d \frac{k}{\varepsilon^3})$. The two operations of computing $\mathbf{C}$ and its top singular value may require $O(d \frac{k^2}{\varepsilon^6})$ time. However, these operations do not necessarily have to occur every iteration if we use the following trick: When entering the while loop we will require the norm of $\mathbf{C} + \mathbf{r}_t \mathbf{r}_t^\top$ to be bounded by $w_t \frac{\varepsilon^2}{k}$ rather than $w_t \frac{\varepsilon^2}{2k}$. Assume now that we entered the while loop at time t . In this case, we do not need to check the condition of the while loop until we reach an iteration t' where $w_{t'} \geq w_t(1 + \frac{\varepsilon^2}{2k})$. Other than checking the condition of the while loop there is no need to compute $\mathbf{C}$, hence the costly operations of the external loop need only be executed an amount of

$$O\left(\log(w_n/w_0) \cdot \frac{k}{\varepsilon^2}\right)$$

We now proceed to analyze the running time of the inner while loop. Each such iteration requires $O(d \frac{k^2}{\varepsilon^6})$ arithmetic operations. However, according to Lemma 4.9 we have that the total number of such iterations is bounded by

$$O\left(\log(w_n/w_0) \cdot \frac{k}{\varepsilon^2}\right)$$

The lemma immediately follows. ■

LEMMA 4.13. (SPACE COMPLEXITY) *Algorithm 1 requires $O(dk/\varepsilon^3)$ space.*

Proof. Immediate from the algorithm and the properties of the Frequent Direction Sketch (Lemma 4.1 item 2) ■

5 Conclusion

Recent discussions with Mark Tygert raise the conjecture that a random projection based PCA technique (see Section 1.1.1) could potentially yield similar, bounds than the ones we achieve here. Another interesting observation is that our algorithm actually bounds $\|\mathbf{X} - \Phi \mathbf{Y}\|_2^2$. To the best of our understating, comparable results are unlikely to be obtainable via random-projection based techniques. Such bounds are useful in the case of noisy data where the optimal registration error is large but the signal is still recoverable. A proof of this fact will appear in the full version of this paper.

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APPENDIX: Omitted Proofs from Section 2

5.1 Proof of Lemma 3.1

Proof. Let $\mathbf{C}_i$ be a value taken by the matrix $\mathbf{C}$ at some point during the algorithm. Let $\ell(i)$ be the number of non-zero vectors in $\mathbf{U}$ at the time of the update to $\mathbf{C}$. For $j \leq \ell$ denote by $\mathbf{U}^{(j)}$ the $d \times \ell$ matrix whose p 'th column is $\mathbf{u}_p$ for $p < j$ and the zero column for $p \geq j$. We prove by induction on i that

1. $\left(\mathbf{U}^{(\ell(i))}\right)^\top \cdot \mathbf{u}_{\ell(i)} = \mathbf{0}_{\ell \times 1}$
2. $\mathbf{C}_i \cdot \left(\sum_{j=1}^{\ell(i)} \lambda_j \mathbf{u}_j \mathbf{u}_j^\top\right) = \mathbf{0}_{d \times d}$

The claim will immediately follow. Notice that since all $\mathbf{u}_j$ are chosen as unit vectors, statement 1 above also indicates that $\mathbf{u}_1, \dots, \mathbf{u}_{\ell(i)}$ form an orthonormal set.

Base case. The base case of the proof is trivial as for $i = 1$ it holds that $\ell(i) = 0$, hence $\mathbf{U}^{(\ell(i))}$ is the all-zeros matrix, and $\mathbf{u}_{\ell(i)} = \mathbf{0}$.

Induction hypothesis. Let $i \geq 1$. We assume the following:

1. $\left(\mathbf{U}^{(\ell(i))}\right)^\top \cdot \mathbf{u}_{\ell(i)} = \mathbf{0}_{\ell \times 1}$

2. $\mathbf{C}_i \cdot \left(\sum_{j=1}^{\ell(i)} \lambda_j \mathbf{u}_j \mathbf{u}_j^\top\right) = \mathbf{0}_{d \times d}$

Induction step. We prove that for $i + 1$:

1. $\left(\mathbf{U}^{(\ell(i+1))}\right)^\top \cdot \mathbf{u}_{\ell(i+1)} = \mathbf{0}_{\ell \times 1}$
2. $\mathbf{C}_{i+1} \cdot \left(\sum_{j=1}^{\ell(i+1)} \lambda_j \mathbf{u}_j \mathbf{u}_j^\top\right) = \mathbf{0}_{d \times d}$

Consider the matrix $\mathbf{C}_{i+1}$. There are two possible reasons for the change in $\mathbf{C}$ and we split the proof to deal with each reason separately.

First reason: $\mathbf{C}_{i+1} = \mathbf{C}_i + \mathbf{r}_t \mathbf{r}_t^\top$ for some t . In this case item 1 holds trivially as $\ell(i) = \ell(i + 1)$. For item 2 (denote $\mathbf{Z} = \left(\mathbf{I}_d - \sum_{j=1}^{\ell(i)} \mathbf{u}_j \mathbf{u}_j^\top\right)$), we have

$$\begin{aligned}
 & \mathbf{C}_{i+1} \cdot \left(\sum_{j=1}^{\ell(i+1)} \lambda_j \mathbf{u}_j \mathbf{u}_j^\top\right) = (\mathbf{C}_i + \mathbf{r}_t \mathbf{r}_t^\top) \cdot \left(\sum_{j=1}^{\ell(i)} \lambda_j \mathbf{u}_j \mathbf{u}_j^\top\right) \\
 & \stackrel{(i)}{=} (\mathbf{r}_t \mathbf{r}_t^\top) \cdot \left(\sum_{j=1}^{\ell(i)} \lambda_j \mathbf{u}_j \mathbf{u}_j^\top\right) \\
 & \stackrel{(ii)}{=} (\mathbf{Z} \mathbf{x}_t \mathbf{x}_t^\top \mathbf{Z}^\top) \cdot \left(\sum_{j=1}^{\ell(i)} \lambda_j \mathbf{u}_j \mathbf{u}_j^\top\right) \\
 & \stackrel{(iii)}{=} \mathbf{0}_{d \times d}
 \end{aligned}$$

Equality (i) is due to induction hypothesis 2. Equality (ii) is due to the definition of $\mathbf{r}_t$ and equality (iii) holds since the induction hypothesis for item 1 indicates that $\mathbf{u}_1, \dots, \mathbf{u}_{\ell(i)}$ are orthonormal, hence $\mathbf{Z} \cdot (\mathbf{I}_d - \mathbf{Z}) = \mathbf{0}_{d \times d}$.

Second reason: $\ell(i + 1) = \ell(i) + 1$ and $\mathbf{C}_{i+1} = \mathbf{C}_i - \lambda_{\ell(i+1)} \mathbf{u}_{\ell(i+1)} \mathbf{u}_{\ell(i+1)}^\top$. Furthermore $\mathbf{u}_{\ell(i+1)}, \lambda_{\ell(i+1)}$ are the top eigen-pair of $\mathbf{C}_i$. For item 1, notice that according to our induction hypothesis (item 2) $\mathbf{C}_i$ is a symmetric matrix whose row and column space are orthogonal to $\mathbf{u}_1, \dots, \mathbf{u}_{\ell(i)}$. Hence, as $\mathbf{u}_{\ell(i)+1}$ is an eigenvector of $\mathbf{C}_i$ it must be the case that $\mathbf{u}_{\ell(i)+1}$ is orthogonal to $\mathbf{u}_1, \dots, \mathbf{u}_{\ell(i)}$.

Next, we proceed to item 2. Denote $\mathbf{Z} = \left(\sum_{j=1}^{\ell(i)+1} \lambda_j \mathbf{u}_j \mathbf{u}_j^\top\right)$. Notice that

$$\begin{aligned}
\mathbf{C}_{i+1} \cdot \left(\sum_{j=1}^{\ell(i+1)} \lambda_j \mathbf{u}_j \mathbf{u}_j^\top \right) &= \\
&= \left(\mathbf{C}_i - \lambda_{\ell(i)+1} \mathbf{u}_{\ell(i)+1} \mathbf{u}_{\ell(i)+1}^\top \right) \cdot \mathbf{Z} \\
&\stackrel{(i)}{=} \mathbf{C}_i \cdot \left(\mathbf{I}_d - \mathbf{u}_{\ell(i)+1} \mathbf{u}_{\ell(i)+1}^\top \right) \cdot \mathbf{Z} \\
&\stackrel{(ii)}{=} \mathbf{C}_i \cdot \left(\mathbf{I}_d - \sum_{j=1}^{\ell(i)} \mathbf{u}_j \mathbf{u}_j^\top \right) \cdot \left(\mathbf{I}_d - \mathbf{u}_{\ell(i)+1} \mathbf{u}_{\ell(i)+1}^\top \right) \cdot \mathbf{Z} \\
&\stackrel{(iii)}{=} \mathbf{C}_i \cdot \left(\mathbf{I}_d - \sum_{j=1}^{\ell(i)+1} \mathbf{u}_j \mathbf{u}_j^\top \right) \cdot \mathbf{Z} \\
&\stackrel{(iv)}{=} \mathbf{0}_{d \times d}
\end{aligned}$$

Equality (i) holds as $\mathbf{u}_{\ell(i)+1}, \lambda_{\ell(i)+1}$ are an eigen-pair of $\mathbf{C}_i$. Equality (ii) is due to induction hypothesis 2. Equality (iii) and (iv) holds as $\mathbf{u}_1, \dots, \mathbf{u}_{\ell(i)+1}$ are an orthonormal set. ■

5.2 Basic Facts First, we show that the matrices $\mathbf{X}, \tilde{\mathbf{X}}$, and $\mathbf{R}$ satisfy some form of the pythagorean theorem for matrices. This result will be useful in a technical manipulation in Lemma 3.5, where we provide an upper bound for $\|\mathbf{R}\|_F^2$.

LEMMA 5.1. *Let $\mathbf{X} \in \mathbb{R}^{d \times n}$, $\tilde{\mathbf{X}} \in \mathbb{R}^{d \times n}$ and $\mathbf{R} \in \mathbb{R}^{d \times n}$ be the matrices whose t 'th column is $\mathbf{x}_t \in \mathbb{R}^d$, $\tilde{\mathbf{x}}_t = \mathbf{U}_t \mathbf{U}_t^\top \mathbf{x}_t \in \mathbb{R}^d$, and $\mathbf{r}_t = \mathbf{x}_t - \tilde{\mathbf{x}}_t \in \mathbb{R}^d$, respectively (the vectors $\mathbf{r}_t$'s are taken as the ones in the end of the corresponding iteration in Algorithm 1). Then,*

$$\|\mathbf{X}\|_F^2 = \|\tilde{\mathbf{X}}\|_F^2 + \|\mathbf{R}\|_F^2.$$

Proof. Recall that for any projection matrix $\mathbf{P}$ and vector $\mathbf{x}$ it holds that $\|\mathbf{x}\|_2^2 = \|\mathbf{P}\mathbf{x}\|_2^2 + \|(\mathbf{I} - \mathbf{P})\mathbf{x}\|_2^2$. According to Lemma 3.1 we have that $\mathbf{P}_t = \mathbf{U}_t \mathbf{U}_t^\top$ is a projection matrix for all t hence,

$$\|\mathbf{x}_t\|_2^2 = \|\mathbf{P}_t \mathbf{x}_t\|_2^2 + \|(\mathbf{I} - \mathbf{P}_t) \mathbf{x}_t\|_2^2 = \|\tilde{\mathbf{x}}_t\|_2^2 + \|\mathbf{r}_t\|_2^2$$

Summing over all t 's yields the claim.

The next two lemmas, provide upper bounds for the dimensions of the subspaces spanned by the columns of $\sum_{j=1}^{\ell'} \lambda_j \mathbf{u}_j \mathbf{u}_j^\top$ and $\mathbf{C}_z$, respectively. We will use those results in Lemma 5.4, where one requires a description of the SVD of the sum $\sum_{j=1}^{\ell'} \lambda_j \mathbf{u}_j \mathbf{u}_j^\top + \mathbf{C}_z$ to calculate a bound for $\|\mathbf{R}\|_2^2$.

LEMMA 5.2. *Let $[\lambda_j, \mathbf{u}_j]$, for $j = 1, \dots, \ell'$ be the pairs of eigenvalues/eigenvectors computed in Algorithm 1 with $[\lambda_j, \mathbf{u}_j]$ computed before $[\lambda_{j+1}, \mathbf{u}_{j+1}]$. Then,*

$$\text{rank} \left(\sum_{j=1}^{\ell'} \lambda_j \mathbf{u}_j \mathbf{u}_j^\top \right) = \ell'.$$

Proof. This result follows immediately from Lemma 3.1 and Lemma 3.3. Specifically, from Lemma 3.1, we have that the $\mathbf{u}_j$'s are orthogonal to each other and from Lemma 3.3 we have that $\lambda_j > 0$, hence $\text{rank}(\sum_{j=1}^{\ell'} \lambda_j \mathbf{u}_j \mathbf{u}_j^\top) = \ell'$.

LEMMA 5.3. *Let $\mathbf{C}_z$ be the matrix $\mathbf{C}$ upon termination in Algorithm 1. Then,*

$$\text{rank}(\mathbf{C}_z) \leq d - \ell'.$$

Proof. We give a proof by contradiction. Assume that $\text{rank}(\mathbf{C}_z) := \rho > d - \ell'$.

For notational convenience, let $\sum_{j=1}^{\ell'} \lambda_j \mathbf{u}_j \mathbf{u}_j^\top := \mathbf{B}$. From Lemma 5.2, we have that $\text{rank}(\mathbf{B}) = \ell'$, hence $\mathbf{B} = \mathbf{U}_B \Sigma_B \mathbf{U}_B^\top$ is the SVD of $\mathbf{B}$ with $\mathbf{U}_B \in \mathbb{R}^{d \times \ell'}$ and $\Sigma_B \in \mathbb{R}^{\ell' \times \ell'}$. From our assumption, we have that $\text{rank}(\mathbf{C}_z) = \rho$, hence $\mathbf{C}_z = \mathbf{U}_{C_z} \Sigma_{C_z} \mathbf{U}_{C_z}^\top$ is the SVD of $\mathbf{C}_z$ with $\mathbf{U}_{C_z} \in \mathbb{R}^{d \times \rho}$ and $\Sigma_{C_z} \in \mathbb{R}^{\rho \times \rho}$. Here, Σ_{C_z} contains strictly positive diagonals. From Lemma 3.1 we have that $\mathbf{C}_z \mathbf{B} = \mathbf{0}_{d \times d}$, which is equivalent to saying that $\mathbf{U}_{C_z}^\top \mathbf{U}_B = \mathbf{0}_{\rho \times \ell'}$. Also, $\mathbf{U}_{C_z}$ contains $\rho > d - \ell'$ columns.

From one hand, based on the relation $\mathbf{U}_{C_z}^\top \mathbf{U}_B = \mathbf{0}_{\rho \times \ell'}$, we have that every column in $\mathbf{U}_{C_z}$ is in $\text{span}(\mathbf{I} - \mathbf{U}_B \mathbf{U}_B^\top)$.

From the other hand, based on the fact that $\mathbf{U}_{C_z}$ contains $\rho > d - \ell'$ orthonormal columns, we have that the last $\rho - d - \ell' \geq 1$ columns in $\mathbf{U}_{C_z}$ are in $\text{span}(\mathbf{I} - \mathbf{U}_{C_z} \mathbf{U}_{C_z}^\top)$.

The two statements contradict each other because, if $\text{span}(\mathbf{I} - \mathbf{U}_B \mathbf{U}_B^\top) = \text{span}(\mathbf{I} - \mathbf{U}_{C_z} \mathbf{U}_{C_z}^\top)$, it would have not been possible that $\mathbf{U}_{C_z}^\top \mathbf{U}_B = \mathbf{0}_{\rho \times \ell'}$.

Finally, we prove a result which says that the spectral norm square of $\mathbf{R}$ is at most that of the max spectral norm of $\mathbf{C}_z$ and the spectral norm of $\sum_{j=1}^{\ell'} \lambda_j \mathbf{u}_j \mathbf{u}_j^\top$. This result is useful in proving Lemma 3.6.

LEMMA 5.4. *Let $\mathbf{R} \in \mathbb{R}^{d \times n}$, be the matrix whose t 'th column is $\mathbf{r}_t \in \mathbb{R}^d$ (the vectors $\mathbf{r}_t$'s are taken as the ones in the end of the corresponding iteration), respectively. Let $\mathbf{C}_z$ be the $d \times d$ matrix $\mathbf{C}$ at the end of the algorithm, and Let $\lambda_j, \mathbf{u}_j$ for $j = 1, \dots, \ell'$ be the eigenvalues, and corresponding eigenvectors, computed in Algorithm 1 with λ_j computed before λ_{j+1} . Then,*

$$\|\mathbf{R}\|_2^2 = \|\mathbf{R}\mathbf{R}^\top\| = \max \left\{ \|\mathbf{C}_z\|, \left\| \sum_{j=1}^{\ell'} \lambda_j \mathbf{u}_j \mathbf{u}_j^\top \right\| \right\}.$$

Proof. For notational convenience, let $\sum_{j=1}^{\ell'} \lambda_j \mathbf{u}_j \mathbf{u}_j^\top := \mathbf{B}$. From Lemma 5.2, we have that $\text{rank}(\mathbf{B}) = \ell'$, hence $\mathbf{B} = \mathbf{U}_B \Sigma_B \mathbf{U}_B^\top$ is the SVD of $\mathbf{B}$ with $\mathbf{U}_B \in \mathbb{R}^{d \times \ell'}$

and $\Sigma_{\mathbf{B}} \in \mathbb{R}^{\ell' \times \ell'}$. From Lemma 5.3, we have that $\text{rank}(\mathbf{C}_z) \leq d - \ell'$, hence $\mathbf{C}_z = \mathbf{U}_{\mathbf{C}_z} \Sigma_{\mathbf{C}_z} \mathbf{U}_{\mathbf{C}_z}^\top$ is the SVD of $\mathbf{C}_z$ with $\mathbf{U}_{\mathbf{C}_z} \in \mathbb{R}^{d \times (d - \ell')}$ and $\Sigma_{\mathbf{C}_z} \in \mathbb{R}^{(d - \ell') \times (d - \ell')}$. Here, $\Sigma_{\mathbf{C}_z}$ might contain some zero diagonals. From Equation 3.4 we have: $\mathbf{R}\mathbf{R}^\top = \mathbf{C}_z + \mathbf{B}$, and from Lemma 3.1 we have that $\mathbf{C}_z \mathbf{B} = \mathbf{0}_{d \times d}$. Hence, for $\mathbf{V}\mathbf{V}^\top = \mathbf{V}^\top \mathbf{V} = \mathbf{I}_d$, we obtain that:

$$\mathbf{R}\mathbf{R}^\top = \mathbf{C}_z + \mathbf{B} = \underbrace{\begin{pmatrix} \mathbf{U}_{\mathbf{C}_z} & \mathbf{U}_{\mathbf{B}} \end{pmatrix}}_{\mathbf{V} \in \mathbb{R}^{d \times d}} \underbrace{\begin{pmatrix} \Sigma_{\mathbf{C}_z} & \\ & \Sigma_{\mathbf{B}} \end{pmatrix}}_{\mathbf{D} \in \mathbb{R}^{d \times d}} \underbrace{\begin{pmatrix} \mathbf{U}_{\mathbf{C}_z}^\top \\ \mathbf{U}_{\mathbf{B}}^\top \end{pmatrix}}_{\mathbf{V}^\top}.$$

Then,

$$\|\mathbf{R}\|_2^2 = \|\mathbf{V}\mathbf{D}\mathbf{V}^\top\|_2 = \|\mathbf{D}\|_2 \leq \max\{\|\Sigma_{\mathbf{C}_z}\|_2, \|\Sigma_{\mathbf{B}}\|_2\}.$$